Not All Cops Are Bayesian: Selection Neglect and Biased Predictions in Policing

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Abstract

Policing is increasingly data-driven, but crime data is a selected sample that reflects where crime is observed, not where it happens. We run a lab-in-the-field experiment testing how senior police officers in Colombia infer crime from selected data. Officers take crime data at face value, turning differences in observability into biased predictions. This failure reflects officers' mental models of crime data, not missing information. Because crime is more observable among minorities, selection neglect can sustain inaccurate statistical discrimination purely from bad inference on selected data.

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I Introduction

Policing is increasingly data-driven, but crime data is a selected sample of crime, weighted by its observability. Consider the Bogotá localities of Barrios Unidos and Tunjuelito. According to crime reports, the official crime data, Barrios Unidos has twice the rate of theft. A data-driven officer who takes this crime data at face value would infer more crime in Barrios Unidos. However, when the residents of these areas are surveyed directly, they report the same incidence of theft. Why then does crime data point to Barrios Unidos? The answer lies in crime observability: a victim of crime in Barrios Unidos is twice as likely to report it to the authorities. Crime data shows where crime is more observable, not where it is more common.

Selection bias in crime data is a general problem for policing. Police departments across the world routinely use citizen reports, calls to 911, or field interviews filed by officers as crime data. All these sources suffer from similar selection bias, caused by reporting behavior or differential enforcement. Crucially, selection bias does not just add noise to crime data: it means crimes are systematically more observable among certain groups. In the United States, crimes committed by minorities are more observable, as they are patrolled, stopped, and searched more often (Chen et al., 2023; National Academies of Sciences, Engineering, and Medicine, 2023; Pierson et al., 2020). Because observability differs across groups, taking crime data at face value would produce systematically biased predictions against the more-observed group: statistical discrimination with no animus behind it. While theory has suggested this source of bias (Che et al., 2024; Hübert & Little, 2023), whether officers actually neglect selection when predicting crime, and with what consequences, remains untested.

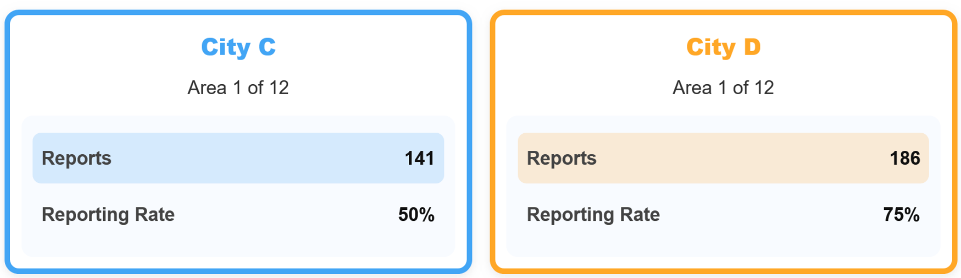
Colombia provides an ideal setting to answer this question: the observability of crime is measurable (the likelihood that victims report a crime), and it is driven by citizen reporting rather than enforcement, making it exogenous to any individual officer's decisions. We leverage national and local victimization surveys to measure the extent of selection bias in crime reports (DANE, 2023; EPV, 2025), documenting that reporting rates vary systematically across types of crime, victim characteristics, and areas. Because of selective reporting, the map of reported crime does not reflect the map of actual victimization. Despite crime reports being selected, official police doctrine instructs officers to use them to assign patrols (PNVCC, 2010). We use proprietary, high-frequency GPS data covering every patrol unit in eight Bogotá *cuadrantes* (police jurisdictions) in June 2025 to assess whether patrols in fact track this selected data. We find they do: one additional crime report in a 300-meter cell is associated with roughly a third more patrol time. Moreover, our participating senior officers report relying on crime reports above any other source. Together, survey and administrative data suggest that officers act on a selected signal.

Although patrolling data shows the co-location of patrols and crime reports, this data cannot tell us whether officers adjust for selection bias, because enforcement, crime, and observability are jointly endogenous. Identifying selection neglect requires exogenous variation in reporting that leaves true crime and enforcement unchanged, which is unlikely to exist in the field: criminals respond to changes in observability (Gonçalves, Jacome, & Weisburst, 2025), and enforcement itself affects observability through monitoring (Chen et al., 2023) and trust (Ang et al., 2025). We therefore create this variation experimentally, eliciting crime predictions from the officers who make them in the field.

In this paper, we conduct a lab-in-the-field experiment with senior officers of the National Police of

Colombia. These officers have on average 14 years of experience, and a quarter of them had just served as station commanders, using crime data to assign patrols. Officers complete a series of crime prediction problems. In each, they see crime data about two fictitious areas and are incentivized to correctly infer in which of them more crimes happened. In particular, officers see the number of *reported crimes* (how many crimes have been observed), along with the *reporting rate* (the probability of observation). Thus, officers can either take the selected crime reports at face value, or easily adjust for selection by dividing them by the reporting rate. Consider the example in Figure 1. An officer who takes reports at face value would infer more crime in the area on the right, but one that adjusts for the difference in observability would make the opposite prediction. By varying reporting rates and true crime independently each round, we generate the identifying variation the field cannot provide. This task not only matches the data available to officers in the field, but is, by design, the simplest version of prediction from selected crime data. Officers are told the data generating process, they are provided with precise information about observability, the adjustment needed is a simple division, and they even have a calculator available on screen. In other words, we design a setting where we can cleanly identify selection neglect, even though we make neglect less likely here than in the field.

Figure 1: Main Experimental Design



Officers neglect selection bias when predicting crime from selected data. Our design generates two cases of predictions. In cases where the area with more reports is also the area with more true crime, and so adjusting for selection does not change the prediction, officers make a mistake in less than a fourth of decisions. This error rate already absorbs any errors unrelated to selection, like inattention. In contrast, officers systematically make the wrong prediction in the cases where adjusting for selection changes the prediction, as in Figure 1. Comparing these two cases, selection neglect causes a 36 percentage point excess in prediction errors. This pattern persists through the experiment, and is not driven by confusion about the task or lack of arithmetic skills: it is robust to focusing on officers who made no mistake on the instructions comprehension questions or on a numeracy test. Crucially, neglect does not only reduce accuracy, it biases predictions towards wherever crime is more observable. Holding crime constant, officers are significantly more likely to infer more crime in the area with the higher reporting rate. By neglecting selection, officers turn differences in observability into biased crime predictions.

These results, based on random variation in reporting rates, also extend to the differences in observability across groups that are widespread in the field. We introduce groups in our setting by bundling areas in two abstract cities, ruling out prejudice. We implement two conditions. In the baseline, reporting rates are identically distributed in both cities, so there is no selection bias across groups. In the treatment condition, reporting rates are systematically higher in one city, creating selection bias across groups. Importantly, we hold crime equal in both cities and both conditions, making any prediction bias against a group entirely the result of inference error. As expected, officers are equally likely to infer crime in both cities at baseline, when crime is equally observable across them. In contrast, officers in the treatment condition become 20 percentage points more likely to predict crime in the city where crime is more observable, despite crime rates being identical. This evidence is consistent with prior theoretical work (Hübert & Little, 2023) and shows that *inaccurate* statistical discrimination arises from bad inference on selected data, with no need for stereotypes or racial animus. Because minorities are overselected in crime data (Chen et al., 2023; Feigenberg & Miller, 2022; Gonçalves, Mello, & Weisburst, 2025; Vomfell & Stewart, 2021), our results establish selection neglect as a cognitive mechanism that can feed policing disparities.

Why do officers neglect selection even in the most favorable setting? Making the correct crime prediction in this setting involves two steps: (1) correctly *representing* the inference problem, this is, recognizing that crime reports are a selected sample of crime, and (2) conditional on that representation, correctly *computing* the adjustment. Several pieces of evidence point towards neglect being a failure of representation. First, when given the option to view reporting rates or not before making one prediction, over a third of officers choose to ignore it, consistent with this data not being demanded by their mental model of the problem. Importantly, information demand is linked to previous experience in the field: officers responsible for reading crime data and assigning patrols are significantly more likely to look at reporting rates than officers in administrative roles. Second, we tap into mental models of crime data by asking officers a battery of questions about crime data at the end of the experiment. We find that officers who are more aware of the biases of this data *in the field* are better at correcting selection bias *in the lab*. Third, most officers are able to correctly solve an equivalent problem not framed as a crime prediction —calculating the number of students in a class from incomplete observations— but then fail to implement this calculation when part of a crime prediction. This suggests officers bring to the task the mental models about crime data developed in the field. Fourth, making reporting rates more salient does not affect neglect, suggesting mental models are entrenched and not developed ad hoc. In contrast, we find no evidence of computational frictions driving the results. Overall, selection neglect among officers seems a representational failure, as officers bring to the lab incorrect mental models developed in the field.

We further test these mechanisms through a randomized treatment, with the idea that giving officers the correct answer will only help if neglect arises from computation, but will not if it is a representational failure. We randomize officers into three groups. A third receive advice from a *Correcting Algorithm* that adjusts for selection and recommends the correct prediction, another third gets it from a *Neglecting Algorithm* that takes reports at face value —so it often gives wrong advice, and a control group who gets no advice. Officers are told which data their algorithm uses and are free to follow or override its advice. We find a striking pattern: officers override the advice that clashes with the inference rule they applied prior to the algorithm, and follow it when it confirms their predictions. As a result, even an algorithm that correctly computes the adjustment fails to improve crime predictions, but a naive algorithm actively makes them worse. These results, that rule out salience or computational frictions as the drivers of neglect, illustrate the limits of treatments that often work in the lab when applied to experienced practitioners with entrenched mental models.

Related work

Policing increasingly relies on crime data to predict crime and guide policing decisions (Brayne, 2020; Perry et al., 2013). Prior work has addressed the consequences of selection bias in crime data, both for predictive algorithms trained on it (Arnold et al., 2025; Lum & Isaac, 2016) and for analysts trying to make inference from such datasets (Durlauf & Heckman, 2020; Gonçalves, Mello, & Weisburst, 2025; Knox et al., 2020). However, how police officers infer crime from selected crime data remains unexplored. We provide the first belief-level evidence that officers take selected crime data at face value. In this line, we contribute to a broader literature on how agents in the criminal justice system form (inaccurate) beliefs (Angelova et al., 2025; Arnold et al., 2018; Bhuller & Sigstad, 2024).

Because neglect turns differences in observability into biased predictions, it is a cognitive mechanism for statistical discrimination in policing. This paper joins recent work on the cognitive drivers of policing decisions, like reflection (Dube et al., 2024; Owens et al., 2018), implicit associations (Glaser, 2015), or cognitive depletion (Ferrazares, 2025). Selection neglect and these mechanisms are distinct from taste and stereotypes: neglect generates inaccurate statistical discrimination from selected data on abstract groups. We provide direct evidence that senior officers hold *inaccurate* beliefs even in simple settings. This finding questions outcome tests that benchmark racial bias against accurate beliefs (Anwar & Fang, 2006; Knowles et al., 2001), as formalized by Bohren et al. (2023).

Finally, this paper contributes to a broader literature identifying selection neglect. While some of this work comes from the field (Cruces et al., 2013; DellaVigna & Kaplan, 2007), most prior studies provide evidence from the lab (Ali et al., 2021; Enke, 2020; Esponda & Vespa, 2018; Jin et al., 2021). We complement these approaches by implementing a controlled environment with expert practitioners in the field. This matters for two reasons. The first one is external validity: we document that not only experimental subjects but experts show this cognitive failure, as Koehler and Mercer (2009) and Malmendier and Shanthikumar (2007) do with investors. The second one is arguably more important for research: practitioners have stable mental models, developed with experience, that lab subjects lack. We probe these mental models and find suggestive evidence that they drive selection neglect, as officers' mental representation of the inference problem determines the information they attend to and the computations they make (Bordalo et al., 2026; Esponda et al., 2024; Schwartzstein, 2014). As a consequence of these entrenched representations, interventions that are effective in the lab may not transfer to practitioners.

The rest of the paper proceeds as follows. Section II uses patrolling data to show that Colombian police officers use crime reports to assign patrols, documents the selection bias in this data, and discusses the limits of administrative data. Section III describes our sample of officers and the experimental design. Section IV presents the results on the prevalence of selection neglect, as well as its consequences for statistical discrimination. Section V explores the mechanisms driving neglect, using algorithmic advice to tell them apart. Section VI discusses the implications of our findings and their position within the prior literature and concludes.

II Setting

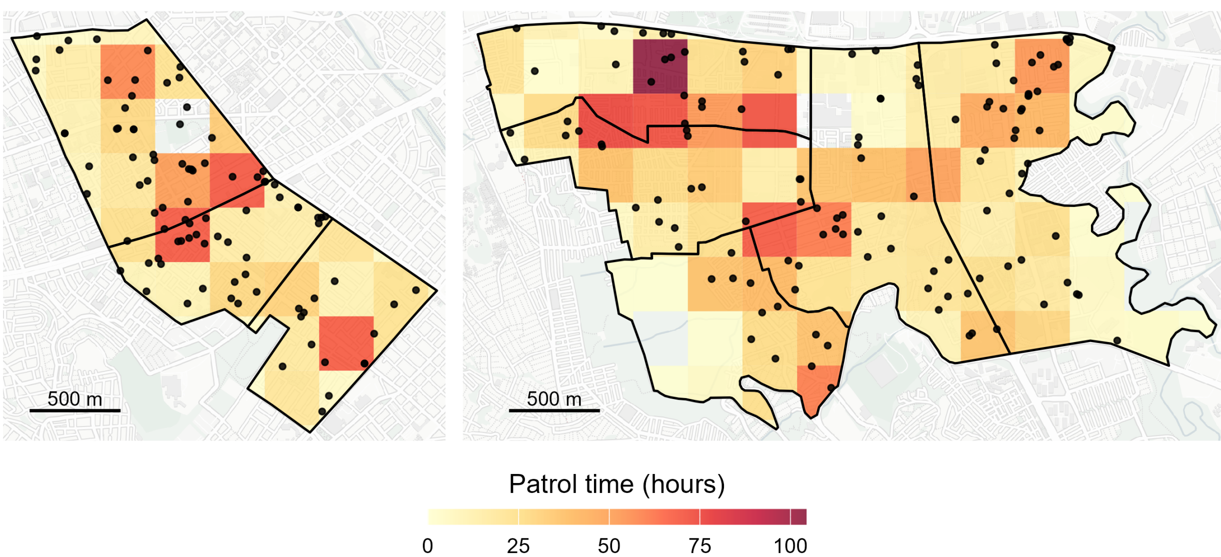
A Data-Driven Policing in Colombia

Policing in Colombia is explicitly data-driven. Crime reports have been a central input into policing decisions for decades, and since the early 2000s they have been consolidated in a centralized statistical system (SIEDCO). Since 2010, patrolling has been governed by the Plan Nacional de Vigilancia Comunitaria por Cuadrantes, which instructs commanders to use reported crime data to decide where and when police are deployed (PNVCC, 2010). Our own survey evidence suggests that officers have adapted to this data-driven regime: among our sample of senior officers, crime reports are considered the most relevant source of information to plan patrols and assign resources, above personal experience, live information from the ground, or even direct orders.

We can see this in administrative patrol data. We obtained the GPS records of every patrol unit assigned to eight police *cuadrantes* in southern Bogotá, for June 2025, and matched them to the crime reports SIEDCO registered in the same area over the same period. We divide the jurisdictions into 300-meter cells and measure how long officers spent inside each one. The result is 150 cells, 218 reported crimes, and an average of seventeen hours of patrol per cell over the month. Figure 2 shows the distribution of crime reports and patrol time over the study area.

Figure 2: Patrol time and crime reports in eight *cuadrantes* of Bogotá, June 2025

(a) CAI Centenario Olaya **(b)** CAI Ismael Perdomo



Notes: Each tile is a 300m × 300m grid cell, shaded by the total time patrol units spent inside it during June 2025, measured from GPS pings weighted by the seconds elapsed until that officer's next fix (capped at 300 seconds). Black outlines are the official *cuadrantes*, the patrol beats to which officers are permanently assigned: three in Centenario and five in Perdomo. Black dots are individual crimes reported to SIEDCO (93 in Centenario, 127 in Perdomo). Tiles are clipped to the cuadrante boundaries for legibility, so the figure displays 123 of the 150 cells used in the analysis. Cells inside a boundary with no shading were never entered by a patrol unit during June, or contain the CAI station building and are excluded throughout. Basemap by OpenStreetMap and Carto contributors.

Patrols co-locate with reported crime. Table A.1 shows that cells with one additional reported crime in June received about six more hours of patrol that month, roughly a third more than the average cell ($p < 0.001$). The pattern holds at higher frequency: one additional report in the previous week is associated with about twenty extra minutes of patrol ($p < 0.001$). These comparisons hold the *cuadrante* fixed, so they compare cells patrolled by the same officers, who are permanently assigned to their *cuadrante*. The concentration of patrol time on reported crime is therefore not the only product of commanders sending more officers to busier areas: it is how officers distribute their time within the area they are responsible for.

These three sources of evidence —official strategy, survey responses, and patrol GPS data— establish crime reports as the central input into policing decisions. But this metric captures only the crimes that are reported. While reports can also be filed by police officers or prosecutors, in practice over 80% are filed by citizens (CEJ, 2025), either in person at a police station or through a dedicated online platform. Because reports are overwhelmingly citizen-filed, which crimes become observable to the police depends on citizen behavior. Not all victims file a report, and the National Police itself acknowledges that reported crime offers only a partial view of underlying crime (Rodríguez Cuadrado, 2008). The central challenge, however, is not that crime reports capture only a subset of crimes: it is that they are not a *representative* sample of crime.

B Selection Bias in Crime Reports

According to national victimization surveys, only 28.8% of crimes are reported to the police (DANE, 2023). Reported crimes are not a random sample of all crimes: there are systematic differences in which crimes are reported and which victims report them. The 2025 Perception and Victimization Survey of Bogotá (EPV, 2025) provides rich data on reporting conditional on victimization. For reference, among over twenty thousand respondents, almost three thousand report being the victims of at least one crime, for a total of 4,403 incidents. Table A.2 regresses whether each of these incidents was reported to the police on victim characteristics and crime type. Reporting varies sharply with both. Younger, higher-income, and more-educated victims are significantly more likely to report. Reporting also varies by crime type. Victims of theft — the most common crime — report in 43% of cases, whereas victims of sexual violence only do so in 27% of cases. Overall, not all crimes are equally observable, creating selection bias in crime data.

Reporting rates vary across space with the composition of crime types and victims, making crime more observable in some areas than others. As a result, the map of reported crime does not correspond to the map of crime. Consider the Bogotá localities of Barrios Unidos and El Tunjuelito, whose theft data is depicted in Figure A.1. According to the EPV, which is representative at the locality level and measures victimization, the incidence of theft is essentially equal across the two localities. Yet *reported* thefts —both according to the EPV and to administrative police data— are twice as high in Barrios Unidos. This asymmetry between crime reports and incidence is driven by reporting behavior: residents of Barrios Unidos are almost twice as likely to report a theft. Therefore, an officer who takes crime reports at face value might confound higher observability with higher incidence: crime data shows where crime is more *observable*, not where it is more *common*.

Importantly for this paper, selection bias in crime data in Colombia is quite transparent and not directly attributable to officers' short-run decisions. Selective reporting determines which crimes become observable, and this behavior is measurable through surveys of residents. Moreover, because officers do not determine the selection themselves, their inference is not confounded by motivated reasoning about their own conduct, as it might be where selection arises from officers' own stop or arrest decisions (Gonçalves, Mello, & Weisburst, 2025; Knox et al., 2020). These features make Colombia an unusually well-suited setting to study whether officers account for selection bias when making crime predictions.

C Do Officers Adjust for Selection Bias?

We have established that the National Police treats reported crime as its primary metric of crime, and this measure suffers of selection bias. Is this selection bias adjusted for when making crime predictions? The *cuadrante* doctrine doesn't consider the observability of crime, and the crime data systems of the National Police do not adjust for differential reporting rates. However, this does not settle whether officers themselves adjust when making crime predictions. Officers know their jurisdictions well, and may understand that some areas or some crimes generate fewer reports than others. On the other hand, they may not adjust because they lack the information to do so: reporting rates are not part of the police statistical system. The question is therefore open: does selection bias in crime reports translate into biased crime predictions, or do officers correct for it? And if officers do not correct for it, is it because they do not have the necessary information?

Why an Experiment

Despite Colombia being an ideal setting to test whether officers neglect selection bias in crime data, this mechanism can't be identified from administrative patrol data. The main reason is the endogeneity of policing, crime incidence, and crime observability. Consider an analysis showing that policing decisions respond to changes in reporting behavior. To cleanly attribute this effect to selection neglect, we would need exogenous variation in the reporting of crime that left crime incidence unchanged. In practice, such variation is unlikely to exist, as criminals respond fast to changes in observability (Gonçalves, Jacome, & Weisburst, 2025; Weisburd, 2021). Moreover, policing decisions often determine observability, both through direct monitoring when patrolling (Chen et al., 2023) and through trust in the authorities (Ang et al., 2025). Finally, because the counterfactual of no policing is never observed and the utility function of police departments is unknown (see Stashko (2023)), we can't determine the accuracy of crime predictions and policing decisions in the field. For these reasons, we need an environment where the payoff function is clear and where crime incidence and observability vary exogenously. Our experimental design puts senior police officers, the relevant decision makers, in such a setting.

III Design

A Participants

Our participants are captains of the National Police of Colombia. Captains usually serve as station commanders, reading crime data to predict crime across space and allocate patrols. In other words, these are the officers responsible for adjusting for selection bias, so they are the relevant sample to study selection neglect. In contrast, patrolling officers follow the orders of station commanders, and officers above captains work in strategic and operational decisions that do not involve direct allocation of patrols.

Procedures. — In September 2025, 202 Captains of the National Police of Colombia were attending the Police Postgraduate School (ESPOL) in Bogotá, completing the mandatory training required for promotion to Major. These officers constitute the universe of captains being promoted in 2025 and were instructed by superior officers to attend our experimental session as part of their training activities. Upon arrival, officers were invited to participate in a "Policing Science Study" and were offered an incentive of $5 for participation, and up to $14 in performance bonus. Eighteen officers (8.9%) declined to participate before receiving any more information about the experiment. Of the 184 officers who began the study, all but two completed it, leaving a final sample of 182 participants and no evidence of endogenous attrition. Officers were paid $10.42 for 33 minutes of experiment, on average.

Final sample. — The final sample consists of 182 captains with substantial professional experience. Officers have an average of 14 years of service in the National Police of Colombia (minimum 12 years), and all hold at least an undergraduate degree, a requirement for promotion to this rank; 15% also hold a master's degree. Importantly for our purposes, participants have extensive experience making operational decisions: 25% were serving as station commanders immediately prior to the training, and an additional 40% held operative positions involving the allocation of police resources. In terms of demographics, 83% of participants are men, and ages range from 32 to 45 years (mean 36). Because our sample consists of experienced captains who are being promoted, it is, if anything, positively selected in terms of ability and experience.

B The Crime Prediction Task

The Crime Prediction Task is the core task in the experiment. In each decision, officers see two abstract *areas* and predict which one had more crime. A number of crimes occurred in each area, which officers never observe [1]. Officers receive $5 if the prediction randomly selected for payment is correct. Note that areas are abstract and carry no social meaning: they are identified only by a label and a color, and each belongs to one of two abstract *cities*. Cities serve as stand-ins for any groups whose crime might differ —age, socioeconomic status, race— without invoking any of them. Because individuals and groups are abstract, officers hold no priors or stereotypes about them, and the design rules out taste-based discrimination by construction.

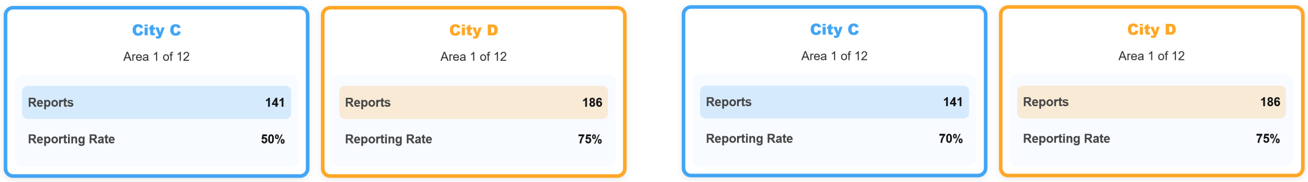
To predict which area had more crime, officers see two pieces of information. The first is the number of crime reports, the crime data used in the field. The second is the reporting rate: the probability that a crime in that area is reported, which measures the observability of crime. Given the data generating

[1] The data generating process of the number of crimes is calibrated to field data, so officers do not see unfamiliar figures. See Appendix XXXX for more details of the DGP.

process, the Bayesian posterior of the number of crimes is the number of reports *adjusted* by the reporting rate: $\frac{\#\text{Reports}}{\text{Reporting Rate}}$. We make sure officers understand how crime reports are generated through an extensive instruction period [2], including multiple understanding questions.

Figure 3: Screens of the Crime Prediction Task

(a) Selection-Sensitive Prediction **(b)** Selection-Neutral Prediction



Notes: The figure shows two examples of decision screens from the Crime Prediction Task. Panel (a) shows a selection-sensitive prediction, where adjusting for the observability of crime (reports/rate) changes the prediction. Panel (b) shows a selection-neutral prediction, where adjusting for observability or taking crime reports at face value lead to the same prediction.

Figure 3 illustrates the decision screen. Consider Panel (a): the area on the left hand side received 141 reports, while the area on the right received 186. Taking reports at face value means predicting higher crime in the area on the right. However, crime reports suffer from selection bias: crime is more observable in the area on the right. Adjusting for observability, the correct answer is that the area on the left had more expected crime ($141/0.5 = 282$) than the area on the right ($186/0.75 = 248$). Because adjusting for selection changes the optimal decision, these predictions identify who neglects selection. Note that adjusting for selection is deliberately simple. The optimal prediction requires one division, officers have an on-screen calculator and one minute per decision, and they complete comprehension and incentivized arithmetic questions before beginning.

Two features of the data-generating process make prediction errors interpretable. First, crime and reporting rates are drawn independently, and officers are told so explicitly. This matters: if officers believed that high-reporting areas also had more crime, following raw reports would be defensible inference rather than error. Independence removes that argument. Second, we fix the number of reports at exactly the reporting rate times the number of crimes, eliminating sampling noise. An officer who adjusts for selection is therefore always correct, so prediction errors reflect cognitive error rather than bad luck.

This design delivers the variation the field cannot provide: crime and observability (reporting rates) move independently across decisions, so we can change how selected the data is while holding true crime fixed. That variation generates two types of predictions, represented in Figure 3. Panel (a) shows a *selection-sensitive* decision, where adjusting for selection changes the consequent prediction: an officer who takes reports at face value necessarily predicts wrong. Panel (b) shows a *selection-neutral* decision, where the area with more reports is also the area with more crime, so adjusting for selection does not change the answer. The two are visually indistinguishable ex ante; telling them apart requires performing the adjustment. We take selection-neutral predictions as a benchmark, as they capture errors unrelated to failing to adjust for selection: inattention, confusion, randomizing, etc. The excess error rate in selection-sensitive decisions is our measure of selection neglect.

[2] Instructions include examples such as: "Imagine that 200 crimes occurred in a area. If the reporting rate is 50%, we expect 100 crimes to be reported."

C Experimental Flow

Introduction — After listening to the instructions, officers complete several training rounds and answer nine comprehension questions. They then answer two incentivized questions that mirror the adjustment needed in the task, but presented as an arithmetic task. Before starting the task, participants can ask any clarifying questions.

Part I — At the beginning of the task, participants complete one *sanity check*, where they are only shown crime reports. In the absence of any other information, participants should predict the area with higher number of reports. We use this decision to filter out inattentive subjects, who comprise 16% of the total sample. Then, all participants complete eight *low-salience* crime predictions, which include three decoy variables in addition to the number of reports and reporting rate [3]. We use these eight predictions to test whether making reporting rates less salient changes predictions. Finally, all participants choose whether to see reporting rates or not for one prediction. We take this *information choice* as evidence of whether they take reporting rates as an input in their inference process, but we do not use the corresponding prediction. One prediction from Part I is randomly chosen for payment.

Main Task – Part II — Participants complete six rounds of the Crime Prediction Task with exactly the same setting described in Figure 3. Across these rounds, crime and reporting rates vary independently and within-subjects. These six rounds are the object of our main analysis on the prevalence of selection neglect among officers.

In these six predictions, we randomize participants into two between-subjects treatments. One half sees *Selection Bias Across Groups*: areas belonging to one city have higher reporting rates on average. The other half sees *No Selection Bias Across Groups*: both cities have the same distribution of reporting rates. Crucially, crime is equally distributed across cities in both treatments. We use these treatments to test whether officers carry selection bias across groups into biased predictions.

Finally, for the last six predictions, we randomize participants into three groups. The *Control* group continues making predictions as in the main task. The *Correcting Algorithm* group is provided with an "algorithmic recommendation" that correctly adjusts for selection bias; participants are informed about this adjustment. The *Neglecting Algorithm* group is provided with a similar algorithmic recommendation, yet this one takes reports at face value and does not adjust for selection bias; participants are informed about this lack of adjustment. This between-subjects treatment serves to test the mechanisms behind neglect, and provides evidence about the limits of information provision. One prediction from Part II is randomly chosen for payment. In total, participants complete 22 crime predictions, out of which we use 20 for our analysis.

End Questions — After completing the experiment, participants answer a short, unincentivized survey. Among these questions, officers report what sources of information are the most relevant for assigning patrols and allocating resources, as well as how aware they are about bias in the crime data they routinely work with in the field. In particular, we ask them whether crime reports capture spatial and temporal crime patterns, whether they are the best available data, whether citizens know how to report crimes, and

[3] In particular, we include the average age, the unemployment rate, and the average socioeconomic status of each area. These variables do not provide any additional information conditional on the number of reports and reporting rates, so they act as uninformative decoys that make reporting rates less salient.

whether reported data is generally biased. Because these responses are highly correlated, we follow Kling et al. (2007) and aggregate z-scores of each into an index of *Bias Awareness*, where higher values reflect a stronger belief of biased crime data on the field. This measure allows us to relate experimental inference errors to officers' beliefs about real crime data. Participants also report their years of experience in the National Police of Colombia and the nature of their most recent position (administrative, operational, or surveillance). Officers in surveillance positions are those acting as station commanders and directly assigning patrols. Finally, participants provide basic demographic information.

Table 1: Summary of Experimental Design

Intro	Comprehension questions; arithmetic questions (up to \$4)
Part I	Ten predictions (1 chosen for \$5 if correct)
Prediction 1	Sanity check (prediction not used for analysis)
Predictions 2-9	Low-salience predictions (decoy information)
Prediction 10	Information choice (prediction not used for analysis)
Part II	Twelve crime predictions (1 chosen for \$5 if correct)
Predictions 1-6	Main Task
Predictions 7-12	Control, Correcting Algorithm, or Neglecting Algorithm
End	Survey questions and demographics

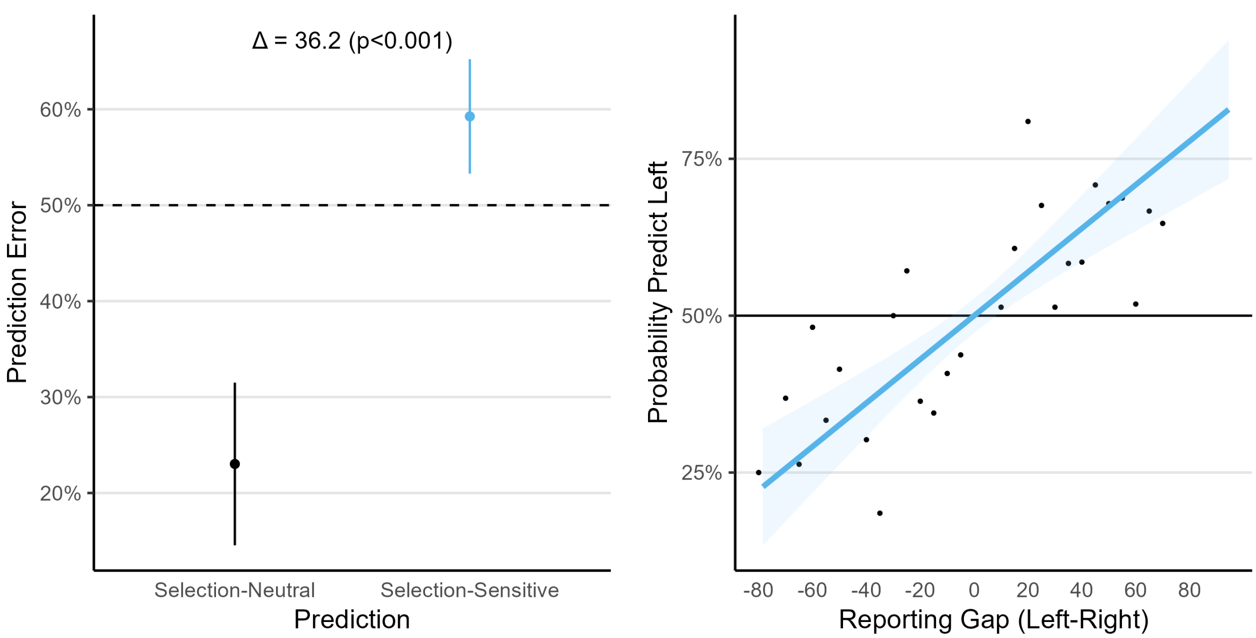
IV Selection Neglect and Biased Predictions

A Officers Neglect Selection Bias in Crime Predictions

We first address the question of whether officers neglect selection bias when making crime predictions. To do so, we focus on the rate of prediction error separately for our two types of predictions. Panel (a) of Figure 4 shows that officers make a mistake in less than a quarter of selection-neutral predictions, where adjusting for selection is not needed to give a correct answer. In contrast, error rises to 59% in selection-sensitive predictions, where the correct answer requires adjusting for selection. Thus, selection neglect causes a 36 percentage point excess in prediction errors. Table A.3 regresses errors on whether the predictions were selection-neutral or selection-sensitive, and finds the selection-driven excess prediction error is highly significant ($p < 0.001$) and robust throughout the experiment.

Figure 4: Evidence of Selection Neglect

(a) Prediction Error by Type of Prediction **(b)** Neglect Turns Selection Bias into Prediction Bias



Notes: Panel (a) shows the average prediction error by prediction type. Selection-neutral predictions, where adjusting for selection does not change the optimal prediction, are depicted in black and in the left column; selection-sensitive predictions, where adjusting for selection does change the optimal prediction, are depicted in blue and in the right column. Error bars represent the 95% confidence interval of the mean, with standard errors clustered at the officer level. The annotation shows the difference across types of predictions, along with the p-value from a t-test clustered at the officer level (see Column (4) of Table A.3). The dashed line shows the random benchmark. Panel (b) plots the probability of predicting higher crime in the area on the left by the difference in reporting rates between areas. Points show the empirical prediction probability at each reporting rate. The blue line shows the fitted linear regression between the two variables, holding crime fixed (see Column (5) of Table A.5). The shaded area shows the 95% confidence interval of the estimate, with standard errors clustered at the officer level. The solid horizontal line shows the optimal prediction: holding crime fixed, reporting rates are uninformative. Observations for both panels are officer-prediction pairs (N = 155 officers × 6 predictions = 930), from the first six predictions of Part II.

Selection neglect is not driven by a handful of officers: it is a widespread error. Figure A.4 shows the distribution of errors at the officer level, divided by selection-neutral and sensitive decisions. While the modal officer makes no mistakes when adjusting for selection isn't needed, they fail *all* predictions that are selection-sensitive. Focusing on the median officer reveals a similar picture: no mistakes in selection-neutral predictions, but wrong predictions in two thirds of sensitive ones.

Neglect does not only reduce accuracy: it biases predictions toward wherever crime is more observable. Crime data shows where crime is more observable, not where it is more common, and an officer who takes reports at face value mistakes the one for the other. We test this by regressing officers' predictions on the reporting-rate gap between the two areas, holding the true crime gap fixed. Panel (b) of Figure 4 plots the probability of predicting higher crime in the area presented on the left as a function of the gap in reporting rates between areas. The figure confirms that officers are significantly more likely to predict higher crime in the area with the higher reporting rate, despite both areas having the same crime. Table A.5 shows this pattern persists across both parts of the experiment and is robust to officer fixed effects.

Selection neglect turns selection bias in crime data into prediction bias. To illustrate its implications, we apply the estimated relationship to the field. Remember the previous example of Barrios Unidos and El Tunjuelito. The difference in reporting rates alone would make an officer roughly 8 percentage points more likely to predict higher crime in Barrios Unidos — not because more crime occurs there, but because more of it is reported. And this is most likely a lower bound on the effect of this cognitive error. Officers in our experiment are provided all the information needed to adjust for selection bias, yet they still mistake observability for crime. In the field, where reporting rates are not included in police statistical systems, this error might be larger.

So far we have shown that neglect makes biases crime predictions. But this error might be innocuous if it only occurred in close calls, where the two areas differ little in true crime. Figure A.5 tests this by disaggregating prediction errors by the crime gap between areas. The opposite holds: the excess prediction error caused by neglect grows with the crime gap. Selection-neutral predictions become easier as the gap widens, while selection-sensitive predictions do not. Neglect therefore misleads officers most in precisely the decisions where being wrong is most costly.

Robustness

Are our results driven by selective effort? — One might be concerned that prediction errors are not driven by selection neglect *per se*, but by officers avoiding a costly calculation in selection-sensitive decisions. We find no evidence to support this. First, the only way to distinguish between neutral and sensitive decisions is computing the selection adjustment, as these decisions are visually identical. Second, officers spend the same time in both types of prediction ($p = 0.437$). Third, we only find a stronger excess prediction error when restricting the sample to those who spend on average more time on selection-sensitive decisions. Overall, our results do not seem driven by less effort in selection-sensitive decisions.

Are our results driven by confusion about the task? — No, selection neglect is not explained by confusion about the task. We went to great lengths to explain the task, and officers complete nine comprehension questions. 32% of officers get all of them right, while two thirds make one or two mistakes. We again regress prediction error on whether the decision is selection-sensitive, but now adding an interaction with whether the officer got all comprehension questions right. Table A.7 finds that selection neglect is robust among officers who answered all comprehension questions correctly. This is not surprising, as selection-neutral predictions should already absorb errors caused by confusion about the task.

Are our results driven by limited arithmetic skills? — No, Table A.7 finds significant neglect among officers who correctly answer all arithmetic questions. This is expected, as the only calculation needed is a division and participating officers can use a calculator on screen.

B Neglect and Bias Across Groups

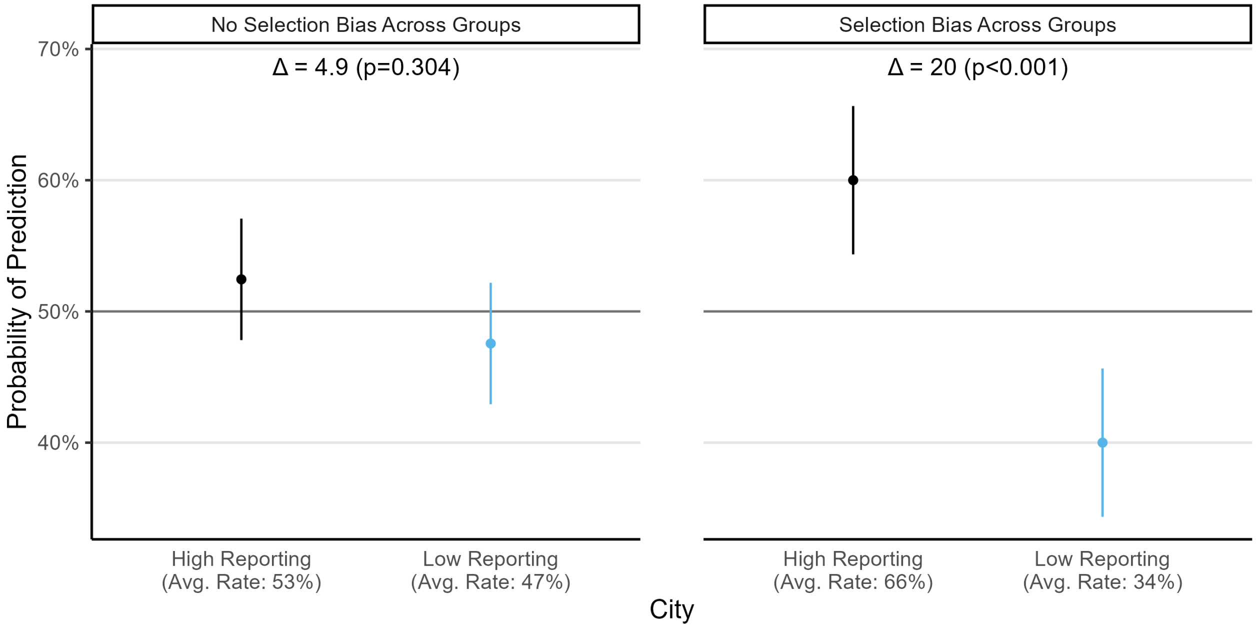
We have used random variation in crime observability to show that area-level differences in data selection translate directly into prediction error, as officers mistake observed crime for true crime. In practice, selection does not vary randomly: there are systematic differences in observability across groups. In Colombia, crimes are more observable when they target a younger, more educated, and higher-SES victim. In the United States, crime is more observable among Black and Hispanic individuals, who are more likely to be patrolled, stopped, searched, reported in a field interview, and arrested (Chen et al., 2023; Gonçalves, Mello, & Weisburst, 2025; National Academies of Sciences, Engineering, and Medicine, 2023). If officers neglect that crime data oversamples one group, officers will predict higher crime among the group where crime is more observable. In other words, neglect can turn differences in crime observability into biased predictions.

We directly test this implication by inducing systematic differences in observability across *cities* in our experiment. Areas within a same city, which acts as an abstract group, share the same distribution of crime; participants are aware of this. In the baseline *No Selection Bias Across Groups*, observability of crime is identical across cities: reporting rates are drawn from the same distribution. In contrast, in the *Selection Bias Across Groups* treatment, reporting rates in one city are 30 percentage points higher on average. Importantly, underlying crime distributions remain identical across cities. Thus, in this treatment cities systematically differ in the number of crimes reported, but this difference is entirely driven by the gap in observability. Under this design, systematic prediction bias across cities can be solely attributed to selection neglect. We randomize officers either into the Baseline or the Selection Bias Across Groups condition for all twelve decisions of Part II. In this section, we focus on the first six of these predictions, as the last six also include the algorithmic recommendation treatments.

Do officers adjust for selection bias across groups, or do they carry it into biased beliefs? Because crime is identical across cities, a rational decision maker should be equally likely to predict higher crime in either city. In contrast, a neglecting decision maker would confound observability with crime, and thus predict higher crime where it is more likely to be reported. To test this prediction, we classify, for each participant, which city had a higher reporting rate on average [4]. Figure 5 plots how likely officers are to predict higher crime in each city, divided by panels according to whether there is systematic selection bias across cities. When crime is equally observable in both cities (left panel), officers are equally likely to predict higher crime in either ($p = 0.304$). In contrast, when crime is more observable in one city, officers become 20 percentage points more likely to predict crime in the group where it is more observable ($p < 0.001$). In other words, officers mistake crime being more observable among one group for crime being more prevalent among that group, even when both groups have identical crime levels.

[4] At baseline, both cities have the same reporting rate in expectation. However, because of randomization, one city always has a slightly higher reporting rate on average. We take this city as the city with higher average reporting rates in the No Selection Bias Across Groups condition.

Figure 5: Prediction bias across groups



Notes: This figure shows how likely the average officer is to predict higher crime in each of the two cities. Cities act as abstract groups to which areas belong. We classify each of the two cities, separately for each participant, as High or Low Reporting, according to which had the higher average reporting rate across the first six predictions of Part II. The figure is divided in panels according to whether both cities have the same reporting rates on expectation (No Selection Bias Across Groups condition), or one city had 30 percentage points higher reporting rates on expectation (Selection Bias Across Groups condition). Error bars show the 95% confidence interval of the mean, with standard errors clustered by officer. The horizontal solid line shows the optimal benchmark: because crime is identical across cities, predictions should be symmetrical. The annotations show the difference in prediction probability across cities, along with the p-value of a t-test clustered at the officer level. Observations are officer-prediction pairs (left panel N = 75 officers $\times$ 6 predictions = 450; right panel N = 80 officers $\times$ 6 decisions = 480), from the first six predictions of Part II.

Because crime is identical across cities by construction, the entire prediction bias caused by neglect is inference error, i.e., *inaccurate*. There is no true difference in crime that a higher prediction for the more-observable group could reflect. This distinction matters for how we interpret discrimination. Tests for discrimination often pit taste-based discrimination against accurate statistical discrimination: if an officer's behavior cannot be explained by rational expectations, it is attributed to animus (Anwar & Fang, 2006; Knowles et al., 2001). Our results show a third possibility. Officers in this task hold no animus, as groups are abstract, and yet their predictions are inaccurately biased against one group. Selection neglect produces statistical discrimination that is simply inaccurate, even though it seems data driven.

This disparity in crime predictions requires strikingly little. There are no stereotypes, no group-specific preferences, and no true difference in crime across groups. A single inference error, taking selected data at face value, is enough. You do not need a biased officer to get biased predictions: bad inference is enough. In the field, where observability differs across real groups — socioeconomic status or race — the same mechanism would push predictions against whichever group is more heavily represented in the data. We return to what this implies for disparities in policing in Section VI.

V Mechanisms

A Framework

We now delineate a simple conceptual framework to build intuition about the possible mechanisms of selection neglect. In our task, making the correct crime prediction requires two steps. First, an officer must correctly *represent* the inference problem, this is, recognize that crime data is a selected sample that is not representative of underlying crime. Second, conditional on this representation, the officer must *adjust* correctly for selection. Neglect can thus arise from a failure at either of these steps, representation or adjustment.

A key marker of a failure of representation is information demand. If an officer does not incorporate selection bias into their inference process, they will not demand this information when making crime predictions. We directly elicit this demand by letting officers choose whether to see reporting rates or not in one crime prediction. There are two non-exclusive explanations for representing the inference problem without considering selection bias: bottom-up and top-down inattention. Salience might drive bottom-up attention to other features of crime data at the expense of selection (Bordalo et al., 2026). On the other hand, officers might hold mental models derived from their experience with this inference problem, which they routinely tackle in the field (Schwartzstein, 2014). If their mental model of crime data does not include its selection, they will not be attentive to this information. We test both sources of inattention in our experiment.

Neglect can also arise at the *adjustment* step, either while making the computation itself or from officers anticipating the complexity of the adjustment and thus opting to apply a simpler heuristic (Arrieta & Nielsen, 2024). While these explanations are unlikely in the Crime Prediction Task, which only requires a simple division, we provide a battery of evidence against neglect being driven by failures at the adjustment step.

B Neglect as a representational failure

We begin by examining the first mechanism: representation of the inference problem. A first marker of this failure is information demand: an officer who chooses not to see reporting rates —the information needed to adjust for selection bias— is not using it as input in their inference. We test this implication directly, letting officers choose whether to see reporting rates for the last decision of Part I. Figure A.6 shows that 36% of officers choose not to see reporting rates for this decision. Moreover, this failure does not only arise in this last prediction. Table A.5 regresses officers' predictions on the difference between areas for every piece of information. We find that officers follow the difference in crime reports to guide their predictions, but, conditional on this difference, they don't change their predictions by the reporting gap between areas. This is especially the case among officers who choose not to see reporting rates in the last prediction of this part, confirming that information demand corresponds to information use across the entire experiment. In other words, officers don't seem to use reporting rates when making crime predictions, and over a third of them do not even demand this information.

Importantly, past experience with crime predictions *in the field* is correlated with information demand. Table A.8 regresses the choice to view reporting rates on instruction comprehension, arithmetic skills, professional experience, and awareness of bias in crime data. We find that ignoring selection is not explained by confusion about the task or lack of basic numeracy. Instead, it varies with officers' previous assignment. Officers whose most recent assignments involved surveillance and patrol assignment (tasks that rely heavily on predicting crime from reported crime data) are 20 percentage points ($p = 0.035$) more likely to view reporting rates than administrative officers. This correlational evidence points to officers forming their mental model about crime prediction and crime data from their experience in the field.

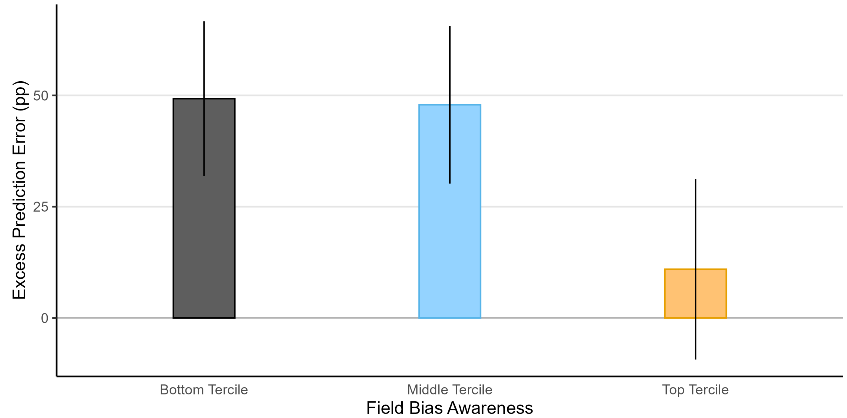
What explains the failure to represent the inference problem correctly? One potential mechanism is salience. In the field, officers might ignore reporting rates because these are not included in official statistical systems, and thus focus instead on more accessible information. While this cannot explain our main results, as we make reporting rates easily available throughout the experiment, we now explore whether their salience is related to neglect. In Part I, reporting rates and crime reports are shown along multiple uninformative variables; in Part II, these decoys are removed, making selection visually unavoidable. Increasing the salience of reporting rates does not make officers incorporate them in their inference process. Figure XXX plots the excess prediction error caused by neglect in both parts of the experiment, and documents similar magnitudes for both salience levels. Column (6) of Table A.6 finds that participants who chose not to see reporting rates, who are a priori the most likely beneficiaries of salience, still fail to use information about selection when made more salient. Overall, the lack of salience of reporting rates does not seem to explain neglect: officers neglect the selective observability of crime even when made more observable.

At the end of the experiment, we ask officers their opinion on five statements about bias in reported crime data, the data they routinely work with. These statements include whether crime reports accurately reflect the (i) spatial and (ii) time trends of crime; (iii) whether citizens know how to report a crime;(iv) whether reported crime data is the best source to predict crime, and (v) whether it provides a biased map of underlying crime. These five questions tap an underlying belief that crime reports misrepresent true crime, and because the answers are highly correlated, we aggregate them into a single index following Kling et al. (2007), reducing measurement error from any single item. We denote this index *field bias awareness*, where higher values reflect greater recognition that reported crime data is a biased sample of crime.

We use the *field bias awareness* index to further test whether officers are applying an incorrect mental model they bring from their work on the field. We divide the sample into terciles of the index and estimate the excess prediction error caused by selection separately for each group. Panel (b) of Figure 6 shows that among the third of officers with the highest awareness, neglect increases prediction error by only 10 percentage points ($p = 0.298$). In contrast, excess prediction error reaches nearly 49 percentage points among the third of the sample with the lowest awareness. In other words, officers who are more aware that reported crime data *in the field* is a biased dataset are able to adjust for selection bias *in the lab*, when given the information to do so. This correlation supports that neglect arises from the wrong mental model of crime data and, together with the previous result on past assignment, suggests these models can develop from experience in the field.

A final piece of evidence comes from the arithmetic questions. We included in the battery a question that closely resembles the Crime Prediction Task, yet with a different frame from crime. In particular, we ask officers "We counted the male students in a class, so we only counted two thirds of the class. If there were 40 men in that class, how many people took that class in total?". This question captures the exact

Figure 6: Selection Neglect by Field Bias Awareness



Notes: The figure shows the excess prediction error caused by selection neglect, estimated separately by terciles of Field Bias Awareness. In particular, we regress prediction error on whether the prediction is selection-sensitive, interacted with a factor variable that divides the sample in terciles. The bars represent the coefficient of selection-sensitivity, along with error bars that show its 95% confidence interval with standard errors clustered at the officer level. Observations are officer-prediction pairs (N = 153 officers × 6 predictions = 918; excludes two officers who did not answer the Field Bias Awareness questions), from the first six predictions of Part II.

adjustment needed in the Crime Prediction Task. When asked in a neutral way, over 75% of officers are able to recover the whole from an observed subsample and get this question right. However, these same officers fail to apply the same adjustment in the Crime Prediction Task. Officers who get this question right still show an excess prediction error of 32 percentage points ($p < 0.001$). Along with the previous results, this suggests that officers are able to perform the necessary computations, but fail to represent the problem correctly when framed as a crime prediction.

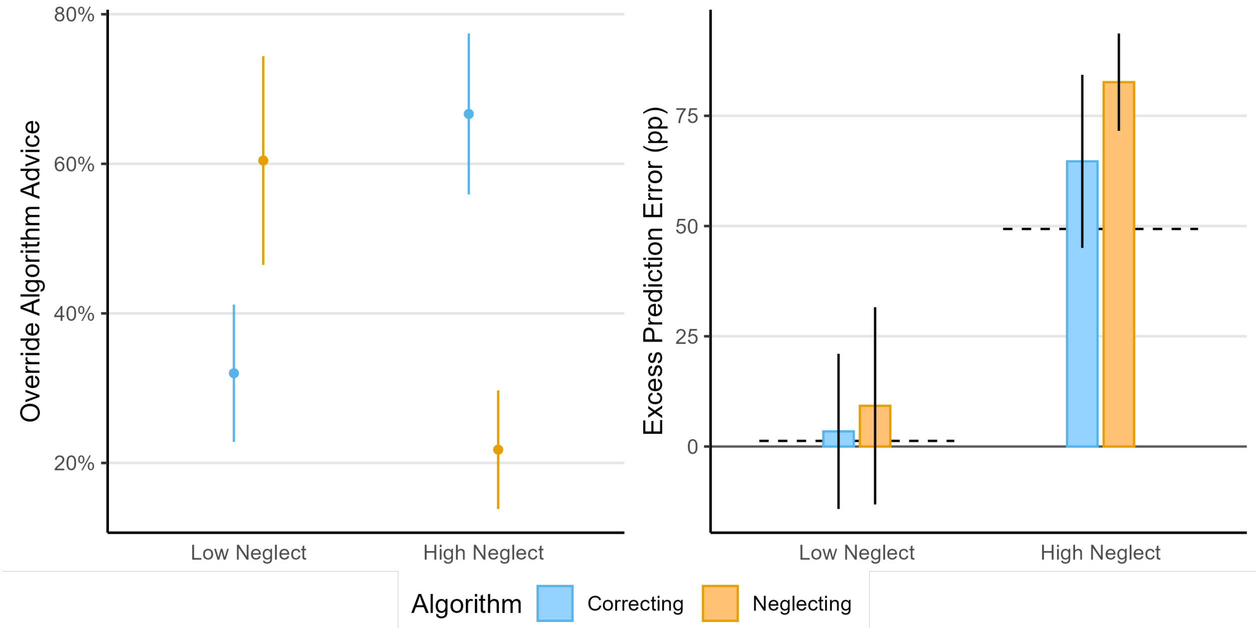
In general, we find little evidence to support the role of computational frictions behind neglect. First, an algorithm that completely eliminates calculation costs does not improve crime predictions nor reduce neglect, as explained in detail in the following subsection. Second, the computation needed is a simple division, and participants have a calculator on screen. However, only 30% of officers ever use it during the main task, and this choice is positively correlated with field bias awareness ($p = 0.024$). Thus, although Table A.3 finds no neglect among participants who used the calculator, we can't rule out this better performance is driven by officers who correctly represent the problem selecting into using the calculator. Overall, the evidence suggests that officers who represent the problem correctly use the calculator to reduce frictions, but these frictions alone cannot account for the excess prediction error caused by selection.

Selection Neglect and Advice Override

We next introduce algorithmic recommendations as a test of the mechanism underlying selection neglect. For the last six predictions of Part II, officers are randomly assigned to one of three conditions: a control group with no recommendation, a correcting algorithm that adjusts reported crime for reporting rates, and a neglecting algorithm that takes reports at face value. Officers are informed whether their algorithm adjusts for reporting rates and retain full discretion to follow or override its recommendations. Because officers retain discretion, adherence and override patterns reveal whether algorithms substitute for inference or are filtered through existing mental models.

Figure 7: Override of algorithm advice and selection neglect

(a) Override in selection-sensitive decisions **(b)** Selection neglect by algorithm treatment



Notes: Panel (a) shows the share of selection-sensitive decisions in which officers override the algorithm recommendation, dividing the sample in half according to their level of neglect in the main analysis. The left column shows officers with Low Neglect —below-median prediction error (< 66%) in selection-sensitive decisions, while the right column shows officers with High Neglect —above-median prediction error. Error bars show the 95% confidence interval of the mean, with standard errors clustered at the officer level. Panel (b) shows the excess prediction error due to selection —the estimated coefficient of selection-sensitivity on prediction error, disaggregated by algorithm intervention and level of neglect. The dashed lines represent the excess prediction error of officers in each group that were randomized to the control algorithm (no algorithm advice). Error bars show the 95% confidence interval of the coefficient, with standard errors clustered at the officer level. Colors show the algorithm treatments: Control in black dashed lines, Correcting in blue, and Neglecting in orange. Observations in both panels are officer-prediction pairs (N = 155 officers × 6 predictions = 930; with 55 officers assigned to Control, 48 to Correcting, and 52 to Neglecting), from the last six predictions of Part II. The left panel subsets this sample to the 601 predictions that are selection-sensitive.

We begin by dividing the sample based on their predictions before the algorithm conditions. Using the first six decisions in Part II, before algorithms are introduced, we classify officers into two groups: those with above the median prediction error in selection-sensitive decisions (high neglect), and those below the median (low neglect). This classification broadly captures whether the inference process of officers was correct before receiving any algorithmic advice.

We first analyze whether officers follow or override recommendations. Panel (a) of Figure 7 reports the fraction of algorithmic recommendations that officers choose to override, focusing on selection-sensitive predictions —where each algorithm gives opposite advice. Officers who show low neglect are twice as likely to override the advice of the neglecting algorithm than the correcting one. In contrast, officers with high neglect display the opposite pattern: they are more than three times as likely to override the correcting algorithm than the neglecting one. In other words, officers tend to ignore the advice that conflicts with their prior behavior. Moreover, Figure A.8 shows that higher field bias awareness is associated with more override of the neglecting algorithm and more adherence to the correcting one. This correlation further supports the interpretation that officers override advice that conflicts with their *representation* of the crime prediction problem.

These adherence and override patterns have clear implications for the effects of the algorithms on selection neglect. Panel (b) of Figure 7 reports excess prediction error by algorithm condition and the prior level of neglect. Among officers with low prior neglect, excess prediction errors remain low across all conditions, as officers reject the advice of the algorithm that takes reports at face value. However, officers with high prior neglect override the corrective advice, slightly increasing neglect with respect to the Control group ($p = 0.264$). In other words, officers fail to adjust for selection bias in crime data even when given the correct answer. These same officers overwhelmingly follow the advice of the Neglecting algorithm, which takes reports at face value, increasing their excess prediction errors by over 30 percentage points ($p = 0.002$). Because officers only follow advice aligned with their preconceived mental model, algorithmic advice fails to correct neglect. Instead, algorithmic advice that does not adjust for selection bias can amplify neglect by validating the wrong model.

The evidence from offering officers algorithmic advice clearly supports that neglect is mostly a representational failure. If neglect was arising from computational frictions, officers should follow the correcting algorithm, eliminating excess prediction errors. Instead, algorithmic advice is filtered through existing mental representations. Information provision that does not change the underlying representation of how crime data are generated is therefore unlikely to eliminate selection neglect, and may even reinforce it.

Four pieces of evidence converge on the same conclusion: selection neglect is a failure of representation, not of information, attention, or computation. First, officers do not demand the information needed to adjust for selection: over a third choose not to see reporting rates, and those who do not look also do not use them when available. Selection is just not part of how they represent the inference problem. Second, this representation tracks officers' experience with crime data in the field. Officers whose assignments involve predicting crime from reports are more likely to consult reporting rates, and officers who recognize the bias of crime data in the field are better at adjusting for selection bias in the lab. The mental model that officers bring from the field predicts how officers use the crime data presented to them. Third, officers who fail to adjust are not unable to compute the adjustment. They correctly perform the same adjustment when it is posed as a neutral arithmetic problem, and they largely ignore an on-screen calculator that would eliminate any computational cost. What they lack is not the capacity to compute the adjustment, but a representation of the inference problem that calls for it. Fourth, the algorithm conditions confirm the representational nature of neglect. If neglect were computational, officers would welcome an algorithm that performs the adjustment for them; instead, high-neglect officers override the correcting algorithm and follow the neglecting one, adopting advice only when it matches the model they already hold. This is the sense in which selection neglect is representational: it is not that officers cannot compute, but that selection does not enter their model of what crime data captures about underlying crime.

VI Discussion

Police departments increasingly rely on crime data to make crime predictions and guide enforcement decisions (Brayne, 2020; Perry et al., 2013). However, crime data often suffers from selection bias: not all crimes are observed, and which depends systematically on the victim, the perpetrator, the type of crime, and where it was committed. This paper shows that senior police officers don't account for selection bias when inferring crime, turning selection bias into biased predictions. We now discuss the broader implications of our findings.

In Colombia, selection bias in crime data comes from selective reporting behavior. The good news is that this bias is measurable through victimization surveys. Addressing the issue of selected data thus requires better measurement of the patterns of selection bias: more frequent and finer-grained victimization surveys that allow a better estimation of reporting rates. At the same time, these surveys offer an alternative metric of crime —victimization—, which can complement the use of selection-adjusted crime reports. However, our results provide a cautionary tale: mental models about crime data are a binding constraint to data-driven policing. We find that simply providing the necessary information for officers to adjust, or even the advice of a perfectly predictive algorithm, does not lead to better predictions, as officers keep their mental model anchored in their past training and experience in the field. Our findings suggest that improving the data ecosystem is not enough: officers must be trained to correctly use this data.

Absent that training, officers will continue to neglect selection bias in crime data. The consequences of neglect for effective and just policing are clear. We show that neglect is *sufficient* to generate biased predictions, absent animus, stereotypes, or true crime differences. We do not address whether this mechanism explains observed disparities in policing, nor directly measure its magnitude in the field, because of the endogeneity of crime and observability in administrative data. However, note that the extent of neglect identified in our experiment is likely to be a lower bound. We create the most favorable conditions for adjusting for selection bias: officers learn the data generating process, have precise information about observability in each area, are incentivized for accuracy, and have no institutional constraints. Despite the favorable conditions, neglect still biases crime predictions. In the field, where none of these conditions hold, neglect is likely to be more pervasive.

The mechanism we identify in this paper is not specific to Colombia or to selection driven by reporting behavior. Anywhere the observability of crime is not complete but also uneven across areas or groups, neglect turns selection bias into biased predictions. Enforcement-driven selection is a specially pernicious case of this mechanism. Police departments across the United States often use police-generated data to make crime predictions, such as crime reports or field interviews (Brayne, 2020; Perry et al., 2013). Because officers disproportionally patrol and stop minority individuals (Chen et al., 2023; Pierson et al., 2020), minorities become overrepresented in crime data, which then justifies further enforcement through higher predicted crime. This feedback loop, described theoretically in detail by Che et al. (2024) and Hübert and Little (2023), is implied by the selection neglect we document among officers whenever selection is endogenous to enforcement decisions. Overall, our findings establish to selection neglect as a cognitive mechanism that can sustain policing disparities.

Our results not only identify a cognitive mechanism for policing disparities, they also question the methods to identify their source. Traditional outcome tests of bias in policing assume that officers make rational predictions from observable data (Anwar & Fang, 2006; Becker, 1971; Knowles et al., 2001). If officers behavior cannot be explained by accurate crime predictions across racial groups, for instance, by accurately anticipating higher criminality among one group, their behavior is attributed to racial animus. However, there is a third possibility: officers might make *inaccurate* crime predictions because of cognitive failures (Bohren et al., 2023). Omitting this possibility can lead analysts to mistake wrong inference for racial bias. By eliciting crime predictions directly from senior police officers, this paper provides direct evidence that officers make *inaccurately* biased predictions about crime. Our results are consistent with Arnold et al. (2018), whose results suggest inaccurate stereotypes drive racial bias in bail decisions. More broadly, our paper contributes to a growing literature that seriously addresses the assumptions of traditional tests, like the selection bias in administrative data (Durlauf & Heckman, 2020; Knox et al., 2020), the ability of officers to accurately target their decisions according to potential outcomes (Feigenberg & Miller, 2022; Rose, 2021), or the actual link between potential outcomes and decisions (Canay et al., 2024; Stashko, 2023). Going forward, analysis of the sources of policing disparities need to take seriously the fact that officers may hold inaccurate beliefs.

That officers are do not make accurate inference from selected data should not be surprising. Adjusting for selection bias in administrative crime data is an issue even for highly trained researchers (Gonçalves, Mello, & Weisburst, 2025) and unbiased predictive algorithms (Arnold et al., 2025), and many lab experiments document the pervasiveness of this failure in more abstract settings (Ali et al., 2021; Araujo et al., 2021; Barron et al., 2024; Enke, 2020; Esponda & Vespa, 2018; Farina & Herman, 2025; Jin et al., 2021). We extend these findings from the lab and identify selection neglect among expert practitioners, who routinely make predictions from selected data in the field. Expert practitioners bring mental models developed with experience that lab participants lack. Our results show that these mental models might not only be the source of cognitive failures, but also a constraint for interventions that address them. Providing officers with the exact data generating process, making selection more salient, or even giving them the correct answer fail to improve predictions, as all these interventions clash with prior mental models. In contrast, similar interventions have been found to work to debias experimental subjects, who lack these field-developed mental models. For instance, Enke (2020) finds that making selection bias more salient curbs neglect, and López-Pérez et al. (2022) find that more transparent and simpler selection processes reduce neglect. In this line, our results warn about the limits of debiasing methods developed in the lab, as they might not transfer to professionals with entrenched representations.

Crime data reflects where crimes are more observable, not just where they are more common. This paper shows that officers take selected crime data at face value, turning biased data into biased predictions. This mechanism requires no animus or stereotypes, yet it is enough to bias officers' predictions and to mislead the tests that attribute such bias to its source. You don't need a biased cop to get biased policing: bad inference is enough.

References

Ali, S. N., Mihm, M., Siga, L., & Tergiman, C. (2021). Adverse and Advantageous Selection in the Laboratory. *American Economic Review*, *111*(7), 2152–2178.

Ang, D., Bencsik, P., Bruhn, J., & Derenoncourt, E. (2025). Community Engagement with Law Enforcement after High-Profile Acts of Police Violence. *American Economic Review: Insights*, *7*(1), 124–142.

Angelova, V., Dobbie, W., & Yang, C. S. (2025). Algorithmic Recommendations and Human Discretion. *The Review of Economic Studies*, rdaf084.

Anwar, S., & Fang, H. (2006). An Alternative Test of Racial Prejudice in Motor Vehicle Searches: Theory and Evidence. *American Economic Review*, *96*(1), 127–151.

Araujo, F. A., Wang, S. W., & Wilson, A. J. (2021). The Times They Are A-Changing: Experimenting with Dynamic Adverse Selection. *American Economic Journal: Microeconomics*, *13*(4), 1–22.

Arnold, D., Dobbie, W., & Hull, P. (2025). Building Nondiscriminatory Algorithms in Selected Data. *American Economic Review: Insights*, *7*(2), 231–249.

Arnold, D., Dobbie, W., & Yang, C. S. (2018). Racial Bias in Bail Decisions. *The Quarterly Journal of Economics*, *133*(4), 1885–1932.

Arrieta, G., & Nielsen, K. (2024). Procedural Decision-Making In The Face Of Complexity.

Barron, K., Huck, S., & Jehiel, P. (2024). Everyday Econometricians: Selection Neglect and Overoptimism When Learning from Others. *American Economic Journal: Microeconomics*, *16*(3), 162–198.

Becker, G. S. (1971, August). *The Economics of Discrimination* (2. edition, Ed.). University of Chicago Press.

Bhuller, M., & Sigstad, H. (2024). Feedback and Learning: The Causal Effects of Reversals on Judicial Decision-Making. *The Review of Economic Studies*, rdae073.

Bohren, J. A., Haggag, K., Imas, A., & Pope, D. G. (2023). Inaccurate Statistical Discrimination: An Identification Problem. *The Review of Economics and Statistics*, 1–45.

Bordalo, P., Conlon, J., Gennaioli, N., Kwon, S., & Shleifer, A. (2026). How People Use Statistics. *The Review of Economic Studies*, *93*(1), 250–285.

Brayne, S. (2020, November). *Predict and Surveil: Data, Discretion, and the Future of Policing*. Oxford University Press.

Canay, I. A., Mogstad, M., & Mountjoy, J. (2024). On the Use of Outcome Tests for Detecting Bias in Decision Making. *Review of Economic Studies*, *91*(4), 2135–2167.

CEJ. (2025). *Informe de estadísticas del SPOA 2025* (tech. rep.).

Che, Y.-K., Kim, J., & Mierendorff, K. (2024, September). Predictive Enforcement.

Chen, M. K., Christensen, K. L., John, E., Owens, E., & Zhuo, Y. (2023). Smartphone Data Reveal Neighborhood-Level Racial Disparities in Police Presence. *The Review of Economics and Statistics*, 1–29.

Conley, T. G. (1999). GMM estimation with cross sectional dependence. *Journal of Econometrics*, *92*(1), 1–45.

Cruces, G., Perez-Truglia, R., & Tetaz, M. (2013). Biased perceptions of income distribution and preferences for redistribution: Evidence from a survey experiment. *Journal of Public Economics*, *98*, 100–112.

DANE. (2023). *Encuesta de convivencia y seguridad ciudadana (ECSC)* (tech. rep.).

DellaVigna, S., & Kaplan, E. (2007). The Fox News Effect: Media Bias and Voting. *The Quarterly Journal of Economics*, *122*(3), 1187–1234.

Dube, O., MacArthur, S. J., & Shah, A. K. (2024). A Cognitive View of Policing. *The Quarterly Journal of Economics*.

Durlauf, S. N., & Heckman, J. J. (2020). An Empirical Analysis of Racial Differences in Police Use of Force: A Comment. *Journal of Political Economy*, *128*(10), 3998–4002.

Enke, B. (2020). What You See Is All There Is. *The Quarterly Journal of Economics*, *135*(3), 1363–1398.

EPV. (2025, November). *Encuesta de Percepción y Victimización de Bogotá 2025* (tech. rep.).

Esponda, I., & Vespa, E. (2018). Endogenous sample selection: A laboratory study. *Quantitative Economics*, *9*(1), 183–216.

Esponda, I., Vespa, E., & Yuksel, S. (2024). Mental Models and Learning: The Case of Base-Rate Neglect. *American Economic Review*, *114*(3), 752–782.

Farina, A., & Herman, C. (2025). Communicating with Data Generating Processes: An Experimental Analysis.

Feigenberg, B., & Miller, C. (2022). Would Eliminating Racial Disparities in Motor Vehicle Searches have Efficiency Costs? *The Quarterly Journal of Economics*, *137*(1), 49–113.

Ferrazares, T. (2025). Shift Structure and Cognitive Depletion: Evidence from Police Officers.

Glaser, J. (2015). *Suspect Race: Causes and Consequences of Racial Profiling*. Oxford University Press.

Gonçalves, F., Jacome, E., & Weisburst, E. (2025). Community Engagement and Public Safety: Evidence from Crime Enforcement Targeting Immigrants.

Gonçalves, F., Mello, S., & Weisburst, E. K. (2025, August). *Selection Bias and Racial Disparities in Police Use of Force* (tech. rep. No. w34175). National Bureau of Economic Research.

Hübert, R., & Little, A. T. (2023). A Behavioural Theory of Discrimination in Policing. *The Economic Journal*, *133*(655), 2828–2843.

Jin, G. Z., Luca, M., & Martin, D. (2021). Is no news (perceived as) bad news? An experimental investigation of information disclosure. *American Economic Journal: Microeconomics*, *13*(2), 141–173.

Kling, J. R., Liebman, J. B., & Katz, L. F. (2007). Experimental Analysis of Neighborhood Effects. *Econometrica*, *75*(1), 83–119.

Knowles, J., Persico, N., & Todd, P. (2001). Racial Bias in Motor Vehicle Searches: Theory and Evidence. *Journal of Political Economy*, *109*(1), 203–229.

Knox, D., Lowe, W., & Mummolo, J. (2020). Administrative Records Mask Racially Biased Policing. *American Political Science Review*, *114*(3), 619–637.

Koehler, J. J., & Mercer, M. (2009). Selection Neglect in Mutual Fund Advertisements. *Management Science*, *55*(7), 1107–1121.

López-Pérez, R., Pintér, Á., & Sánchez-Mangas, R. (2022). Some conditions (not) affecting selection neglect: Evidence from the lab. *Journal of Economic Behavior & Organization*, *195*, 140–157.

Lum, K., & Isaac, W. (2016). To Predict and Serve? *Significance*, *13*(5), 14–19.

Malmendier, U., & Shanthikumar, D. (2007). Are small investors naive about incentives? *Journal of Financial Economics*, *85*(2), 457–489.

National Academies of Sciences, Engineering, and Medicine. (2023, May). *Reducing Racial Inequality in Crime and Justice: Science, Practice, and Policy*. National Academies Press.

Owens, E., Weisburd, D., Amendola, K. L., & Alpert, G. P. (2018). Can You Build a Better Cop? *Criminology & Public Policy*, *17*(1), 41–87.

Perry, W. L., McInnis, B., Price, C. C., Smith, S., & Hollywood, J. S. (2013, September). *Predictive Policing: The Role of Crime Forecasting in Law Enforcement Operations* (tech. rep.).

Pierson, E., Simoiu, C., Overgoor, J., Corbett-Davies, S., Jenson, D., Shoemaker, A., Ramachandran, V., Barghouty, P., Phillips, C., Shroff, R., & Goel, S. (2020). A large-scale analysis of racial disparities in police stops across the United States. *Nature Human Behaviour*, *4*(7), 736–745.

PNVCC. (2010). *Plan Nacional de Vigilancia Comunitaria por Cuadrantes* (tech. rep.).

Rodríguez Cuadrado, J. F. (2008). La historia y el presente de las cifras delictivas y contravencionales en Colombia: Un nuevo conocimiento. *Revista Criminalidad*, *50*(1), 109–118.

Rose, E. K. (2021). Who Gets a Second Chance? Effectiveness and Equity in Supervision of Criminal Offenders. *The Quarterly Journal of Economics*, *136*(2), 1199–1253.

Schwartzstein, J. (2014). Selective Attention and Learning. *Journal of the European Economic Association*, *12*(6), 1423–1452.

Stashko, A. (2023). Do Police Maximize Arrests or Minimize Crime? Evidence from Racial Profiling in U.S. Cities. *Journal of the European Economic Association*, *21*(1), 167–214.

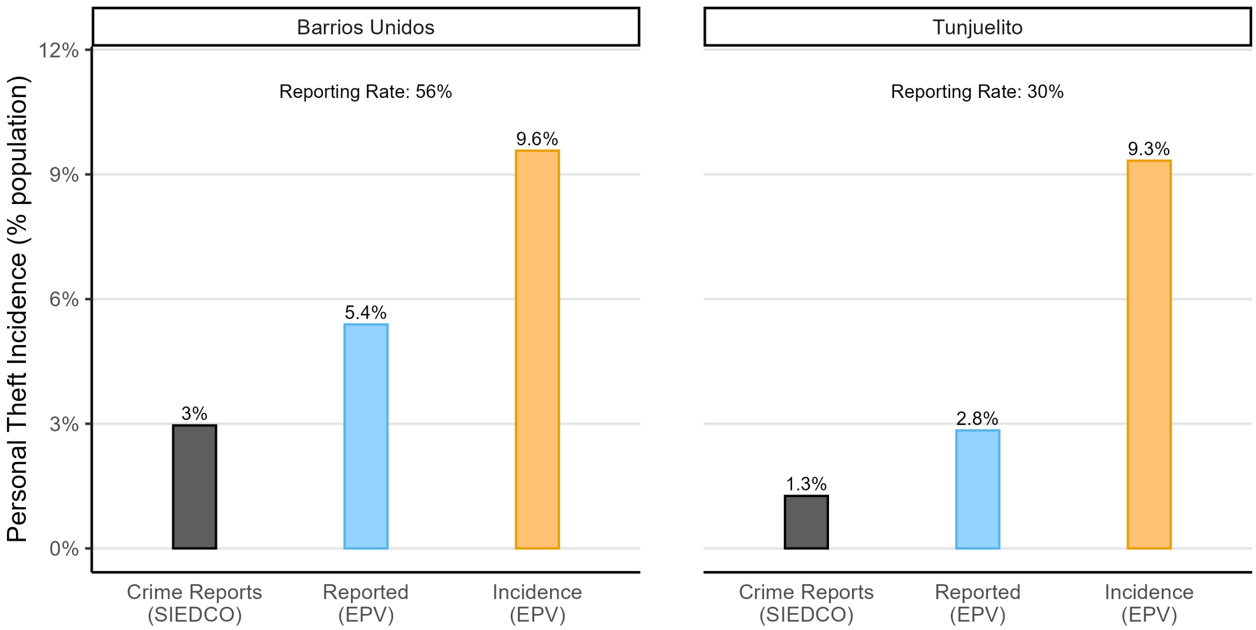
Vomfell, L., & Stewart, N. (2021). Officer bias, over-patrolling and ethnic disparities in stop and search. *Nature Human Behaviour*, *5*(5), 566–575.

Weisburd, S. (2021). Police Presence, Rapid Response Rates, and Crime Prevention. *The Review of Economics and Statistics*, *103*(2), 280–293.

A Appendix

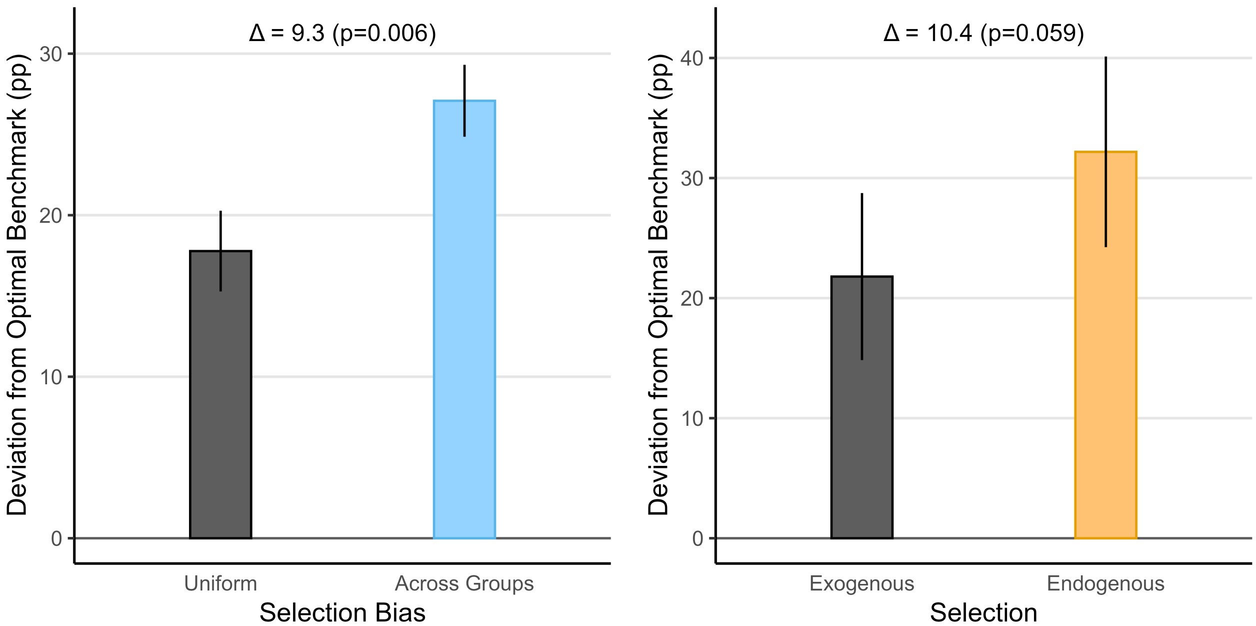
Additional Figures and Tables

Figure A.1: Crime Data on Theft, Barrios Unidos and Tunjuelito 2025



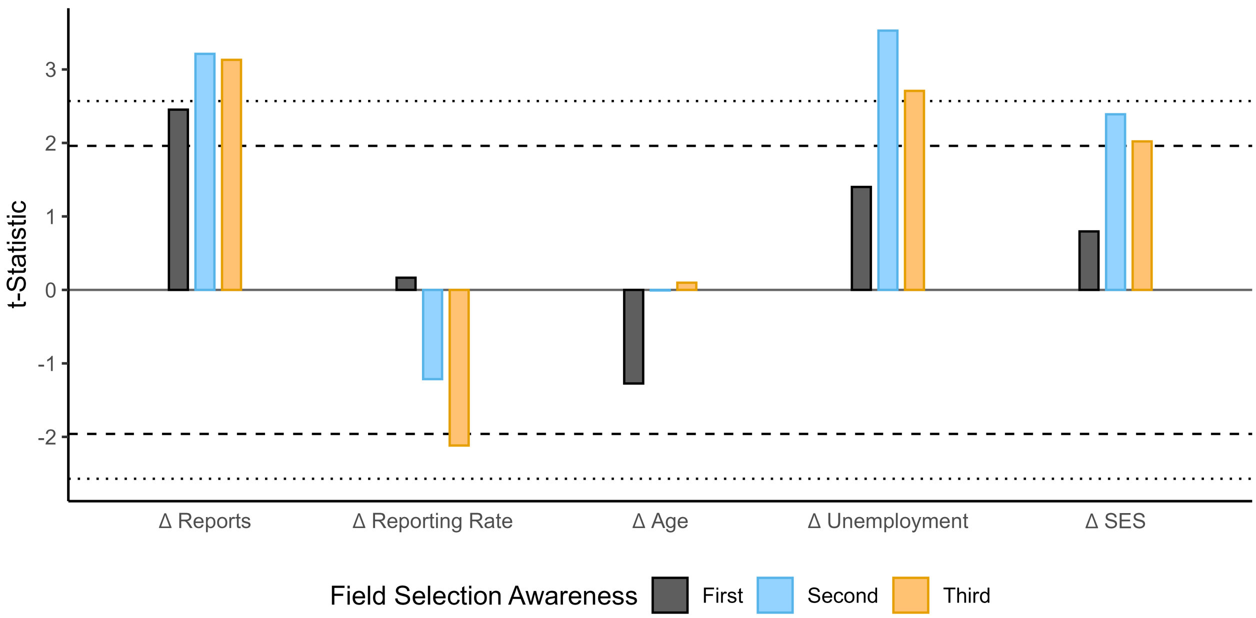
Notes: Personal theft in two Bogotá localities with near-identical survey victimization. Each panel shows three bars, all expressed as a percentage of the locality's population. Crime Reports (SIEDCO) is the count of personal thefts officially reported to the police in the locality during 2025 (SIEDCO administrative data), divided by locality population. Reported (EPV) is the share of respondents who were victimized and say they reported the crime to the police, from the 2025 Encuesta de Percepción y Victimización (EPV). Incidence (EPV) is the share of survey respondents who report having been a victim of personal theft, from the 2025 Encuesta de Percepción y Victimización (EPV); because the EPV is representative at the locality level, this share estimates the victimization rate of the locality's residents. The label above each panel is the reporting rate among residents, Reported (EPV) / Incidence (EPV): the share of victims who reported to police. Victimization is nearly identical across the two localities, yet recorded crime is roughly twice as high in Barrios Unidos — the gap is driven by the reporting behavior, not the amount of crime.

Figure A.2: Prediction Bias Across Groups



Notes: This figure shows prediction bias across groups defined as the deviation from the optimal likelihood to predict higher crime in one of the groups. We first calculate the officer-level optimal benchmark: how often their optimal prediction would be city C. Note that, although on expectation both cities have equal crime, because of randomization the benchmark is not 50% for all officers. We then calculate prediction bias as the deviation from this optimal benchmark, in absolute value, for each officer. Panel (a) shows the prediction bias across Selection Bias conditions. The left column shows the average officer in the *Uniform Selection* treatment, where reporting rates for both cities are equal in expectation. The right column shows officers in the *Selection Bias Across Groups* treatment, where reporting rates are 30 percentage points higher in expectation for one city. The figure only includes the first six predictions of Part II. Panel (b) shows the prediction bias across Endogenous Selection conditions. The left column shows the average officer in the *Exogenous Selection* group, where crime predictions do not affect reporting rates. The right column shows officers in the *Endogenous Selection* treatment, where predicting higher crime in one city increases reporting rates by 15 percentage points in that same city for the next decision. The panel includes the last six predictions of Part II, and only includes officers randomized to see no algorithmic recommendation. In both panels, error bars show the 95% confidence interval of the means, with standard errors clustered by officer. The annotations show the difference between conditions, alongside with the p-value from a t-test clustered at the officer level. The horizontal black line at zero shows the optimal benchmark: because both cities have the same number of crimes in expectation, optimal predictions should be unbiased across cities.

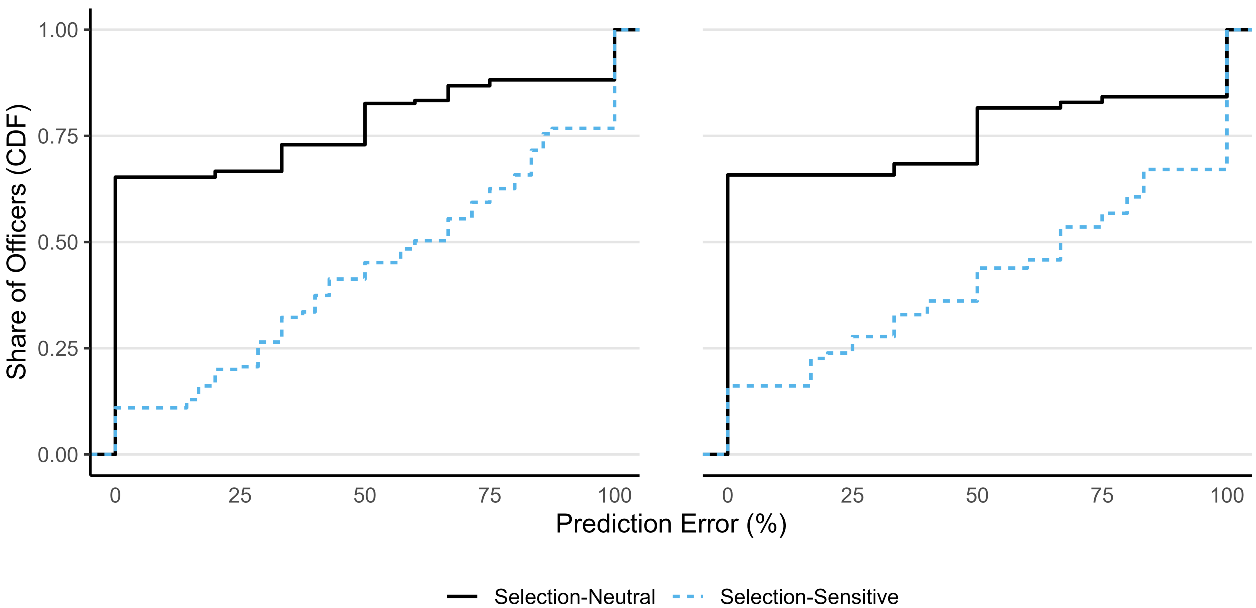
Figure A.3: Predictors of crime predictions, by terciles of field selection awareness



Notes: This figure shows the results of regressing crime predictions on the differences between available variables in Part I, estimated separately by terciles of Field Selection Awareness. The x-axis shows the dependent variables: Δ Reports is the difference in crime reports between the two presented areas at each prediction; Δ Reporting Rate is the difference in reporting rates, in percentage points; Δ Age is the difference in mean age; Δ Unemployment is the difference in unemployment rates, in percentage points; Δ SES is the difference in socioeconomic status (*estrato*, measured from 1 to 6). The y-axis shows the t-statistic of that regressor, estimated separately by tercile of Field Selection Awareness. We use t-statistics to ease comparison between regressors and models. Field Selection Awareness is a composite index measuring how aware officers are about selection bias in the reported crime data they work with in the field. Horizontal dashed lines mark the conventional significance level of 95%, the dotted lines mark the 99% level.

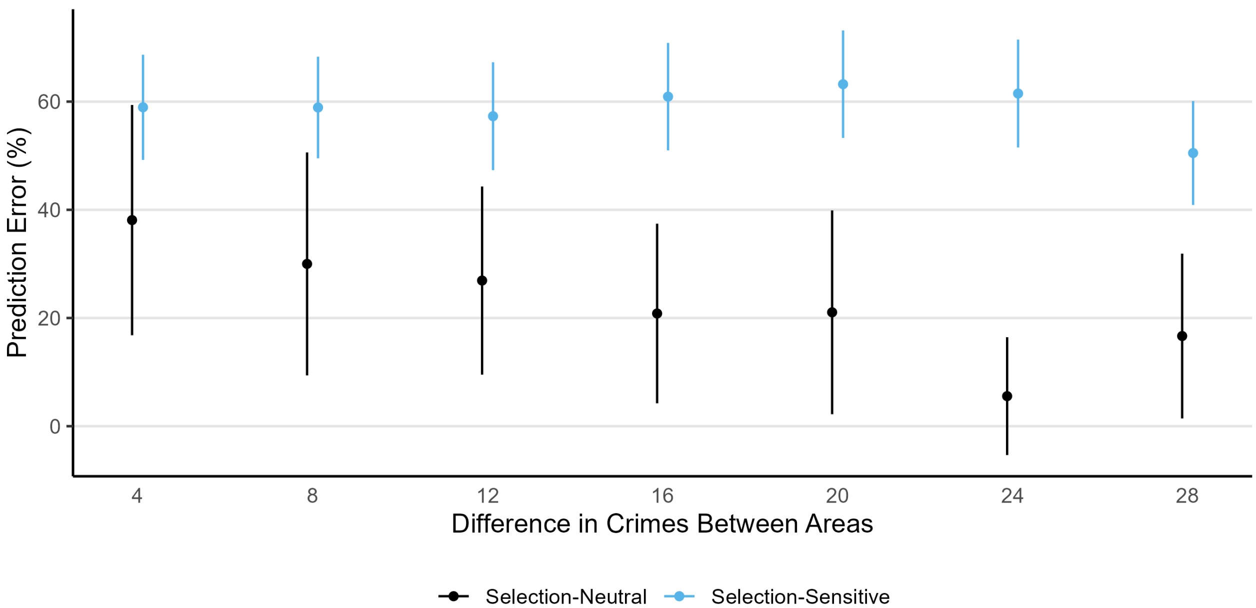
Figure A.4: Distribution of Prediction Error

(a) Part I — Eight Crime Predictions **(b)** Part II — Six Crime Predictions



Notes: This figure shows the cumulative distribution of officer-level prediction error by type of prediction. Both panels divide the data by whether adjusting for selection was needed to make the right prediction. Selection-neutral decisions, where adjusting for selection does not change the optimal prediction, are depicted in black; selection-sensitive decisions, where adjusting for selection does change the optimal prediction, are in blue dashed lines. The left panel uses all eight predictions from Part I, the right panel uses the first six predictions from Part II.

Figure A.5: Prediction Error by Type of Prediction and Difference in Crimes

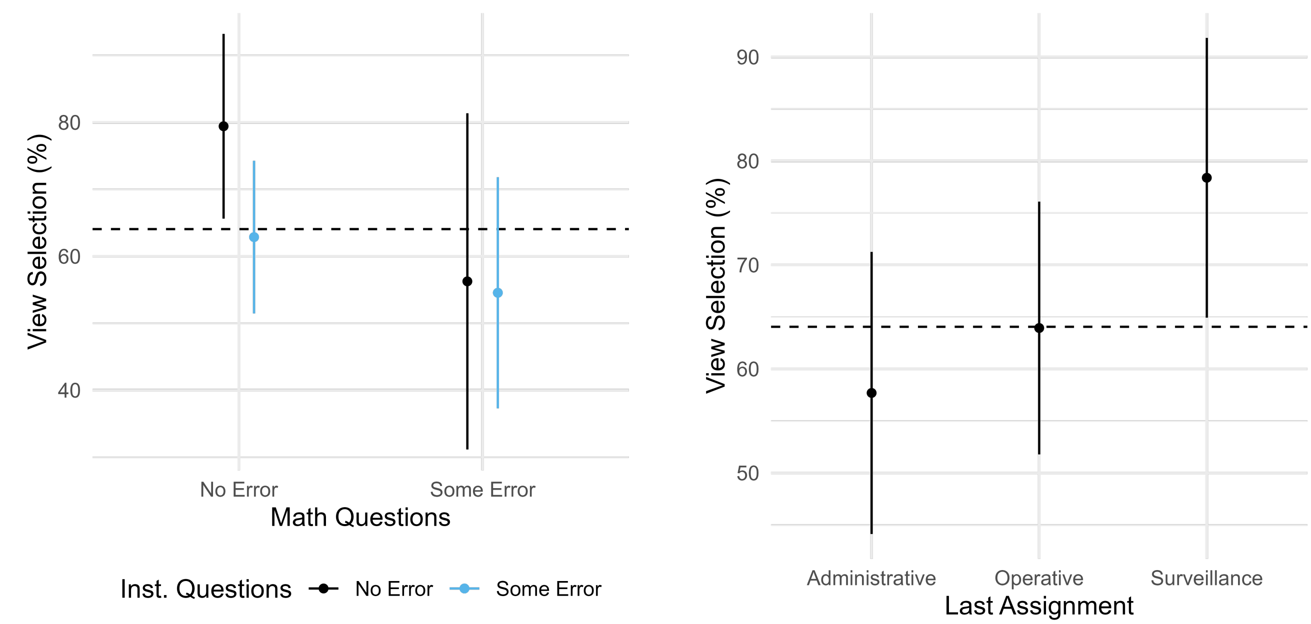


Notes: This figure disaggregates Figure 4 by the difference in crimes between areas at each prediction. The figure further divides the data by whether adjusting for selection was needed to make the accurate prediction. Selection-neutral decisions, where adjusting for selection does not change the optimal prediction, are depicted in black; selection-sensitive decisions, where adjusting for selection does change the optimal prediction, are depicted in blue. Error bars in both panels represent the 95% confidence interval of the mean, with standard errors clustered at the officer level.

Figure A.6: Choice to View Selection

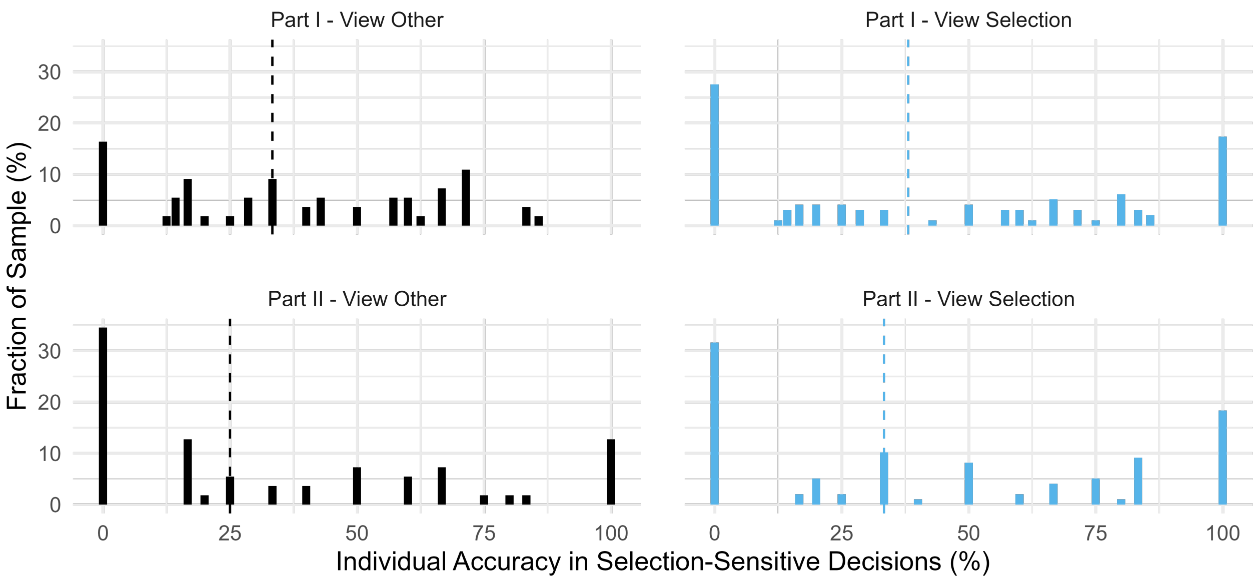
(a) By performance in math and instructions questions

(b) By last assignment



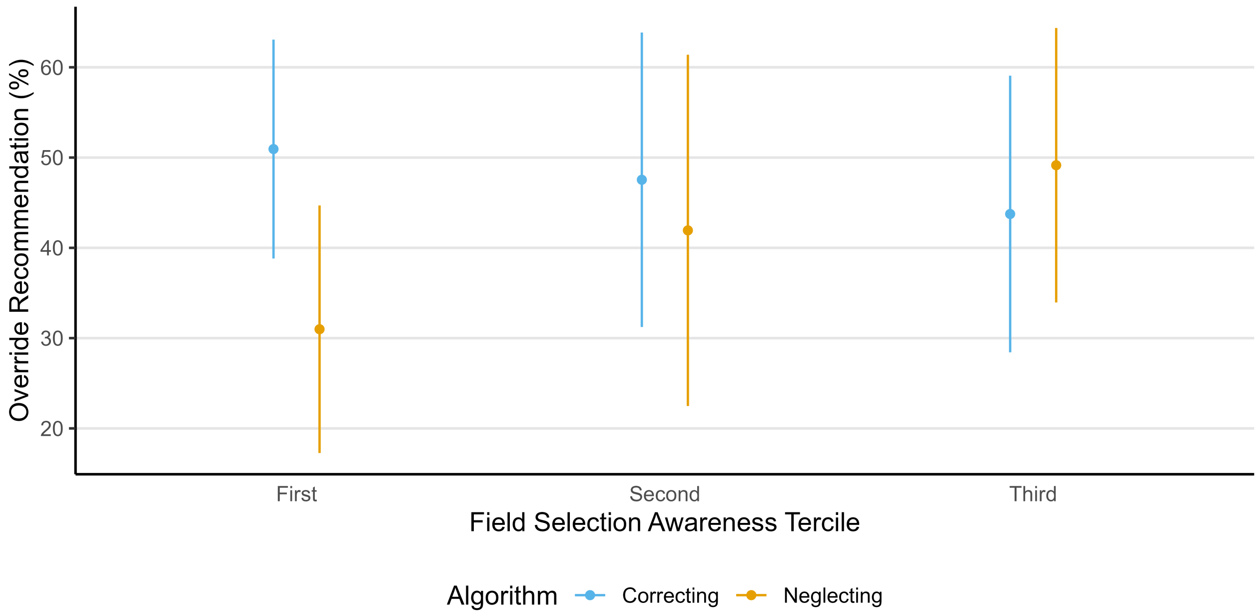
Notes: This figure shows the share of officers who choose to view selection for the last prediction of Part I. Panel (a) divides the sample by performance in the math and instruction comprehension questions. The left column shows officers who answered all math questions correctly, while the right column shows those who made at least one error. In black, we show officers who answered all instruction comprehension questions correctly, while in black we show those who made at least one error. Panel (b) divides the sample by the last assignment of the officer before entering the academy. Note that those with a Vigilance assignment were in charge of predicting crime and assigning patrols. In both panels, error bars represent the 95% confidence interval around the mean. The dashed line shows the sample-level mean at 64%.

Figure A.7: Accuracy Distributions in Selection-Sensitive Decisions



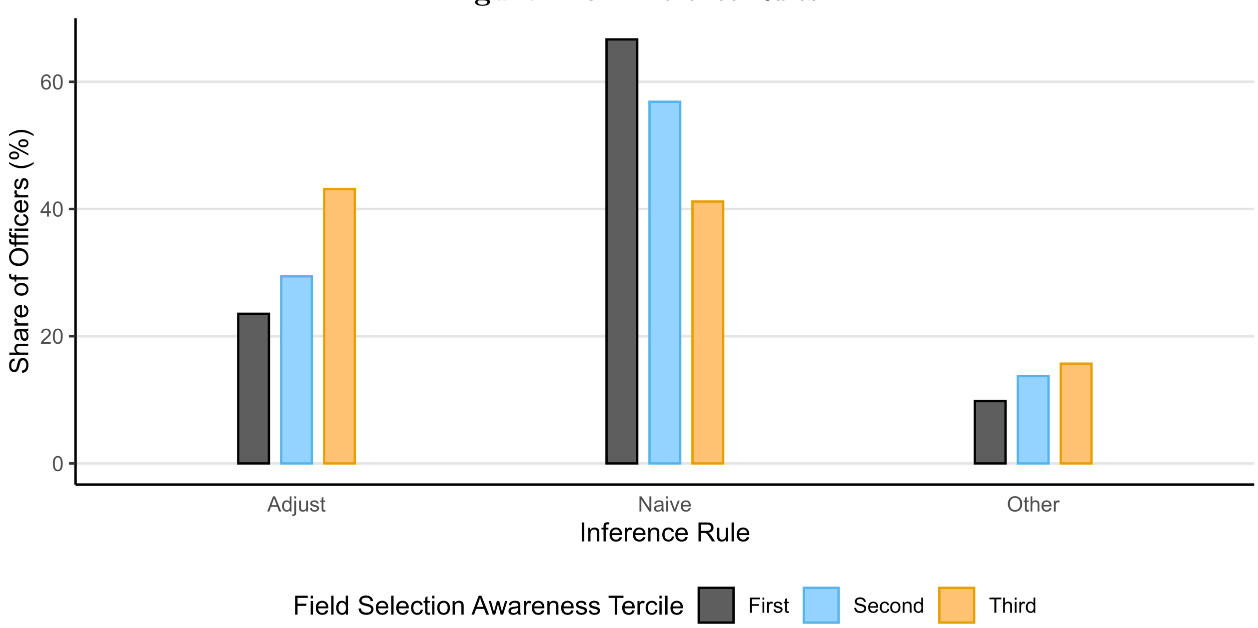
Notes: This figure shows the distribution of officer-level prediction accuracy in selection-sensitive decisions. Plots in the first row show predictions from Part I, where data selection is less salient. Plots in the second row show prediction from Part II, where data selection is made more salient. Columns separate the sample between officers who chose *not* to view the reporting rate for the last prediction of Part I (left, in black), and officers who chose to view selection (right, in blue). When comparing across columns, we notice the strong link between choosing to see selection and prediction accuracy in Part I, where selection wasn't salient. Once selection is made more salient in Part II, officers who hadn't chosen to view it in Part I (left column) do not really improve accuracy but become more likely to systematically neglect it. For officers who already were paying attention to reporting rates, the increase in salience does not seem to have any effect. In both cases, salience does not mitigate neglect. While bottom-up salience increases the likelihood that selection comes to mind, it does not substitute for having the correct adjustment rule.

Figure A.8: Heterogeneity in Advice Override



Notes: This figure shows the share of selection-sensitive decisions in which officers override the algorithm recommendation, dividing the sample in terciles according to their field selection awareness. Error bars show the 95% confidence interval of the mean, with standard errors clustered at the officer level.

Figure A.9: Inference Rules



Notes: This figure shows the share of selection-sensitive decisions in which officers override the algorithm recommendation, dividing the sample in terciles according to their field selection awareness. Error bars show the 95% confidence interval of the mean, with standard errors clustered at the officer level.

Table A.1: Patrolling time and reported crimes across eight police *cuadrantes*

	Patrol Time (sec)				
	Cross-section	Daily panel		Weekly panel	
	(1)	(2)	(3)	(4)	(5)
Reported Crimes (month)	0.308				
	(0.000)				
Reported Crimes, day t-1		0.671	0.015		
		(0.000)	(0.826)		
Reported Crimes, days t-1 to t-7				0.536	0.002
				(0.000)	(0.966)
Fixed-effects					
Cuadrante	Yes	Yes		Yes	
Cell			Yes		Yes
Observations	150	4,350	4,350	3,450	3,427
Dependent variable mean	61,031	2,012	2,012	1,852	1,864

Notes: Results from Poisson pseudo-maximum likelihood (PPML) regressions with $p-$values in parentheses and Conley (1999) spatial standard errors with a 1 km cutoff. The unit of observation is a 300m $\times$ 300m grid cell in column (1) and a cell $\times$ day in columns (2)-(5). The study area is the union of the Centenario Olaya and Ismael Perdomo CAI polygons, buffered by one cell width; the two cells containing the station buildings are excluded throughout. The dependent variable Patrol Time is the time patrolling officers spent in the cell, built from GPS pings weighted by the seconds elapsed until that officer's next fix, capped at 300 seconds. Reported Crimes counts SIEDCO crime reports —including personal injury, theft from persons, vehicle theft, motorcycle theft, and homicide (218 in total). We include Reported Crimes recorded in the cell over the month in column (1), on day $t-1$ in columns (2)-(3), and cumulatively over days $t-1$ to $t-7$ in columns (4)-(5). The panel covers June 2025; columns (2)-(3) drop the first day and columns (4)–(5) the first seven, for which the lags are undefined, and the cell fixed effects in column (5) absorb one cell with no patrol time in the estimation window. *Cuadrante* fixed effects correspond to the eight official patrol beats (three in Centenario, five in Perdomo) to which officers are permanently assigned; Cell fixed effects to the 150 grid cells. Coefficients are semi-elasticities: $\exp(\beta)-1$ is the proportional change in patrol time associated with one additional reported crime.

Table A.2: Correlates of reporting behavior in Bogotá

	Reported to Police
	(1)
Age	-0.002
	(0.001)
SES (Stratum)	0.033
	(0.001)
Female	0.003
	(0.854)
Education	0.021
	(0.000)
Type of crime (ref. Theft - persons)	
Cybercrime	0.068
	(0.005)
Extortion	0.095
	(0.036)
Personal injury	-0.004
	(0.901)
Sexual violence	-0.159
	(0.001)
Theft - bikes	-0.015
	(0.703)
Theft - commerce	-0.099
	(0.017)
Theft - residences	0.057
	(0.121)
Theft - vehicles	0.054
	(0.085)
Threats	-0.023
	(0.545)
Vandalism	-0.118
	(0.002)
Violence against women	-0.030
	(0.479)
Fixed Effects	
Locality	Yes
Occupation	Yes
Observations	4,403
Dependent variable mean	0.435

Notes: Results from a linear regression with $p-$values in parentheses and robust standard errors. The number of observations corresponds to 2,956 respondents who were the victim of a crime, out of 20,038 total respondents, with one row per each crime suffered by the respondent. The dependent variable Reported to Police is a dummy that equals 1 if the respondent reported that crime to the police, conditional on being a victim. Age is measured in years. SES (Stratum) reflects the socio-economic stratum of the respondent, from 1 (High) to 6 (Low). Female is a dummy that equals 1 if the participant identifies as a woman. Education is an ordered factor variable from 1 (None) to 11 (PhD). Type of crime variables are dummies, taking as reference Theft - persons. The model includes Locality (20) $\times$ Occupation (8) fixed effects.

Table A.3: Prediction error across decisions

	Prediction Error					
	Part I			Part II		
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.221	0.327	0.159	0.230	0.333	0.136
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.006)
Selection-Sensitive	0.358	0.282	0.495	0.362	0.323	0.509
	(0.000)	(0.000)	(0.000)	(0.000)	(0.003)	(0.000)
View Selection		-0.165			-0.155	
		(0.007)			(0.111)	
Selection-Sensitive × View Selection		0.119			0.047	
		(0.183)			(0.704)	
Calculator			0.011			0.103
			(0.867)			(0.310)
Selection-Sensitive × Calculator			-0.299			-0.427
			(0.002)			(0.002)
Observations	1,240	1,224	784	930	918	588
Dependent variable mean	49.274	49.346	46.811	53.333	52.832	48.469

Notes: Results from a linear regression with $p-$values in parentheses and standard errors clustered at the officer level. The dependent variable Prediction Error is a dummy that equals 1 if the prediction of which area had more crime was wrong. Selection-Sensitive is a dummy that equals 1 if adjusting for selection is needed to make an accurate prediction. View Selection is a dummy that equals 1 if the officer chose to view the reporting rate for the last decision of Part I. Calculator is a dummy that equals 1 if that officer ever used the on-screen calculator through that part. Columns (1) to (3) use the eight predictions from Part I, with Column (3) restricting the sample to those who chose to view selection in the last decision of Part I. Columns (4) to (6) use all predictions in Part II, excluding the last six predictions for participants randomized into an algorithm treatment other than *Control*. Column (6) restricts the sample to those who chose to view selection in the last decision of Part I.

Table A.4: Neglected crimes across decisions

	Neglected Crimes			
	Part I		Part II	
	(1)	(2)	(3)	(4)
Constant	9.880	10.466	3.026	1.741
	(0.000)	(0.000)	(0.000)	(0.034)
Selection-Sensitive	11.273	-8.718	6.367	-0.997
	(0.000)	(0.001)	(0.000)	(0.285)
Crime Gap		-0.011		0.081
		(0.799)		(0.228)
Selection-Sensitive $\times$ Crime Gap		0.516		0.459
		(0.000)		(0.000)
Observations	1,240	1,240	930	930
Dependent variable mean	18.435	18.435	8.353	8.353

Notes: Results from a linear regression with $p-$values in parentheses and standard errors clustered at the officer level. The dependent variable Neglected Crimes measures the prediction error in each decision: the difference between the highest number of crimes across the two areas and the predicted area. Note that this variable takes a value of zero if the prediction was correct, and the crime gap between areas if the prediction was wrong. Selection-Sensitive is a dummy that equals 1 if adjusting for selection is needed to make an accurate prediction. Crime Gap measures the difference in crimes between the two areas. Columns (1) and (2) use the eight predictions from Part I. Columns (3) and (4) use all predictions in Part II, excluding the last six predictions for participants randomized into an algorithm treatment other than *Control*.

Table A.5: Effect of crime and reporting gaps in prediction probability

	Predict A			Predict C		
	Part I			Part II		
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.469			0.499		
	(0.000)			(0.000)		
Crime Gap	0.002	0.003	0.001	0.003	0.003	-0.001
	(0.000)	(0.000)	(0.098)	(0.011)	(0.007)	(0.674)
Reporting Gap	0.003	0.004	0.003	0.003	0.003	0.003
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.008)
View Selection $\times$ Crime Gap			0.002			0.006
			(0.062)			(0.009)
View Selection $\times$ Reporting Gap			0.000			0.000
			(0.700)			(0.980)
Fixed-effects						
Officer	No	Yes	Yes	No	Yes	Yes
Observations	1,224	1,224	1,224	918	918	918
Dependent variable mean	0.469	0.469	0.469	0.487	0.487	0.487

Notes: Results from a linear regression with $p-$values in parentheses and standard errors clustered at the officer level. The dependent variable Predict A is a dummy that equals 1 if the area from city A was predicted to have higher crime (Columns (1) to (3)). The dependent variable Predict C is a dummy that equals 1 if the area from city C was predicted to have higher crime, using predictions from Part II (Columns (4) to (6)). Crime Gap is the difference in crimes between the two presented areas at each prediction. Reporting Gap is the difference in reporting rates between the two presented areas at each prediction, in percentage points. View Selection is a dummy that equals 1 if the officer chose to view the reporting rate for the last prediction of Part I. Columns (1) to (3) use the eight predictions of Part I, including the gap in demographic decoys as controls. Columns (4) to (6) use the first six predictions in Part II. Columns (2), (3), (5), and (6) include officer fixed effects.

Table A.6: Effect of all observable variables in prediction probability

	Predict A			Predict C		
	Part I			Part II		
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.467			0.494		
	(0.000)			(0.000)		
Δ Crime Reports	0.003	0.005	0.003	0.003	0.004	0.000
	(0.000)	(0.000)	(0.034)	(0.026)	(0.002)	(0.950)
Reporting Gap	-0.003	-0.006	-0.003	-0.005	-0.008	0.004
	(0.088)	(0.002)	(0.325)	(0.178)	(0.026)	(0.434)
Δ Age	-0.001	-0.001	-0.003			
	(0.422)	(0.531)	(0.279)			
Δ Unempt.	0.010	0.008	0.021			
	(0.000)	(0.001)	(0.000)			
Δ SES	0.022	0.017	0.037			
	(0.005)	(0.032)	(0.012)			
View Selection × Δ Crime Reports			0.003			0.007
			(0.105)			(0.008)
View Selection × Reporting Gap			-0.006			-0.018
			(0.159)			(0.008)
View Selection × Δ Age			0.003			
			(0.472)			
View Selection × Δ Unempt.			-0.020			
			(0.000)			
View Selection × Δ SES			-0.031			
			(0.067)			
Fixed-effects						
Officer	No	Yes	Yes	No	Yes	Yes
Observations	1,224	1,224	1,224	918	918	918
Dependent variable mean	0.469	0.469	0.469	0.487	0.487	0.487

Notes: Results from a linear regression with $p-$values in parentheses and standard errors clustered at the officer level. The dependent variable Predict A is a dummy that equals 1 if the area from city A was predicted to have higher crime (Columns (1) to (3)). The dependent variable Predict C is a dummy that equals 1 if the area from city C was predicted to have higher crime, using predictions from Part II (Columns (4) to (6)). Δ Crime Reports is the difference in crime reports between the two presented areas at each prediction. Reporting Gap is the difference in reporting rates between the two presented areas at each prediction, in percentage points. Δ Age is the difference in mean age between the two presented areas at each prediction. Δ Unemployment is the difference in unemployment rates between the two presented areas at each prediction, in percentage points. Δ SES is the difference in socioeconomic status (*estrato*, measured from 1 to 6) between the two presented areas. View Selection is a dummy that equals 1 if the officer chose to view the reporting rate for the last prediction of Part I. Columns (1) to (3) use the eight predictions of Part I. Columns (4) to (6) use the first six predictions in Part II. Columns (2), (3), (5), and (6) include officer fixed effects.

Table A.7: Accuracy and neglected crimes by selection, robustness

	Prediction Error				Missed Crimes			
	Part I		Part II		Part I		Part II	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Constant	0.223	0.177	0.240	0.311	9.637	7.956	3.042	4.089
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.004)
Selection-Sensitive	0.291	0.328	0.304	0.226	8.781	10.424	5.504	4.570
	(0.000)	(0.000)	(0.000)	(0.030)	(0.000)	(0.001)	(0.000)	(0.009)
Some Math Error	-0.006		-0.025		0.683		-0.042	
	(0.921)		(0.790)		(0.800)		(0.978)	
Selection-Sensitive × Some Math Error	0.211		0.178		7.839		2.680	
	(0.020)		(0.166)		(0.052)		(0.195)	
Some Inst. Error		0.070		-0.115		3.093		-1.509
		(0.221)		(0.248)		(0.236)		(0.346)
Selection-Sensitive × Some Inst. Error		0.036		0.197		0.898		2.602
		(0.685)		(0.114)		(0.816)		(0.204)
Observations	1,240	1,240	930	930	1,240	1,240	930	930
Dependent variable mean	0.493	0.493	0.533	0.533	18.435	18.435	8.353	8.353

Notes: Results from a linear regression with $p-$values in parentheses and standard errors clustered at the officer level. The dependent variable Prediction Error is a dummy that equals 1 if the prediction of which area had more crime was wrong (Columns (1) to (4)). The dependent variable Missed Crimes measures the prediction error in each decision: the difference between the highest number of crimes across the two areas and the predicted area (Columns (5) to (8)). Note that this variable takes a value of zero if the prediction was correct, and the crime gap between areas if the prediction was wrong. Selection-Sensitive is a dummy that equals 1 if adjusting for selection is needed to make an accurate prediction. Some Math Error is a dummy that equals 1 if the officer didn't answer all math questions correctly. Some Instruction Error is a dummy that equals 1 if the officer didn't answer all instruction comprehension questions correctly on the first attempt. Columns (1), (2), (5), and (6) use the eight predictions from Part I. Columns (3), (4), (7), and (8) use the first six predictions in Part II.

Table A.8: Heterogeneity in choice to view selection

	View Selection			
	(1)	(2)	(3)	(4)
Constant	0.762	0.577	0.649	0.708
	(0.000)	(0.000)	(0.000)	(0.000)
Some Math Error	-0.132			-0.136
	(0.125)			(0.107)
Some Instruction Error	-0.118			-0.135
	(0.139)			(0.087)
Operative		0.062		0.076
		(0.503)		(0.412)
Surveillance		0.207		0.193
		(0.035)		(0.051)
Reported Data Reliability			0.107	0.110
			(0.212)	(0.181)
Observations	153	150	151	150
Dependent variable mean	0.641	0.653	0.649	0.653

Notes: Results from a linear regression with $p-$values in parentheses and standard errors clustered at the officer level. The dependent variable View Selection is a dummy that equals 1 if the officer chose to view the reporting rates for the tenth decision of Part I. Some Math Error is a dummy that equals 1 if the officer didn't answer all math questions correctly. Some Instruction Error is a dummy that equals 1 if the officer didn't answer all instruction comprehension questions correctly on the first attempt. Operative is a dummy that equals 1 if the officer's last assignment was operative. Surveillance is a dummy that equals 1 if the officer's last assignment was in surveillance —predicting crime and assigning patrols. The reference group is officers whose last assignment was administrative. Reported Data Reliability measures the officer's perception of how reliably reported crime data captures underlying crime, with higher values implying higher reliability. All columns use data at the officer level, as there is only one choice to view selection.

Table A.9: Heterogeneity in accuracy loss due to selection

	Accuracy					
	Part I			Part II		
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.788	0.782	0.788	0.732	0.732	0.754
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Selection-Sensitive	-0.428	-0.358	-0.426	-0.378	-0.338	-0.400
	(0.000)	(0.000)	(0.000)	(0.001)	(0.000)	(0.000)
Operative	0.002		0.007	-0.009		-0.030
	(0.971)		(0.917)	(0.921)		(0.688)
Surveillance	-0.038		-0.050	0.004		-0.034
	(0.609)		(0.465)	(0.972)		(0.706)
Selection-Sensitive × Operative	0.131		0.118	0.102		0.120
	(0.168)		(0.208)	(0.433)		(0.283)
Selection-Sensitive × Surveillance	0.082		0.104	0.015		0.064
	(0.485)		(0.343)	(0.915)		(0.611)
Report Reliability		0.115	0.124		0.278	0.281
		(0.082)	(0.068)		(0.000)	(0.000)
Selection-Sensitive × Report Reliability		-0.255	-0.262		-0.394	-0.406
		(0.010)	(0.008)		(0.000)	(0.000)
Observations	1,216	1,224	1,216	1,224	1,236	1,224
Dependent variable mean	0.512	0.511	0.512	0.467	0.462	0.467

Notes: Results from a linear regression with $p-$values in parentheses and standard errors clustered at the officer level. The dependent variable Accuracy is a dummy that equals 1 if the prediction of which area had more crime was accurate. Selection-Sensitive is a dummy that equals 1 if adjusting for selection is needed to make an accurate prediction. Operative is a dummy that equals 1 if the officer's last assignment was operative. Surveillance is a dummy that equals 1 if the officer's last assignment was in surveillance —predicting crime and assigning patrols. The reference group is officers whose last assignment was administrative. Report Reliability (Reported Data Reliability) measures the officer's perception of how reliably reported crime data captures underlying crime, with higher values implying higher reliability. Columns (1) to (3) use the eight predictions from Part I. Columns (4) to (6) use all predictions in Part II, excluding the last six predictions for participants randomized into an algorithm treatment other than *Control*.

Table A.10: Selection Neglect across Selection Environments

	Accuracy		
	(1)	(2)	(3)
Constant	1.000	0.565	1.000
	(0)	(0.000)	(0.000)
Selection-Sensitive	-0.591	-0.167	-0.600
	(0.000)	(0.260)	(0.000)
Asymmetric	-0.235		-0.455
	(0.000)		(0.000)
Selection-Sensitive × Asymmetric	0.230		0.449
	(0.006)		(0.034)
Endogenous		0.132	-0.345
		(0.324)	(0.003)
Selection-Sensitive × Endogenous		-0.214	0.236
		(0.236)	(0.146)
Endogenous × Asymmetric			0.523
			(0.006)
Selection-Sensitive × Endogenous × Asymmetric			-0.460
			(0.127)
Observations	930	330	330
Dependent variable mean	0.467	0.455	0.455

Notes: Results from a linear regression with $p-$values in parentheses and standard errors clustered at the officer level. The dependent variable Accuracy is a dummy that equals 1 if the prediction of which area had more crime was accurate. Asymmetric is a dummy variable that equals 1 if the officer was randomized into the *Asymmetric Selection* treatment, where areas from one of the two cities have a 30 percentage points higher reporting rate in expectation than the other city. It equals 0 when the officer is in the *Symmetric Selection* treatment, where reporting rates for both cities are drawn from the same uniform distribution, and thus are equal in expectation. Endogenous is a dummy variable that equals 1 if an officer is randomized into the *Endogenous Selection* treatment, where predicting one city to have higher crime increases reporting rates in that city by 15 percentage points in the next prediction. It equals 0 when the officer is in the *Exogenous Selection* treatment, where predictions do not affect future reporting rates. Column (1) uses the first six predictions of Part II. Columns (2) and (3) use the last six predictions in Part II, excluding officers randomized into an algorithm treatment other than *Control*.

Table A.11: Effects of selection environments

	Chose High Rate	Repeat Prediction	Persistent Prediction	
	(1)	(2)	(3)	(4)
Constant	0.496	0.532	0.056	0.000
	(0.000)	(0.000)	(0.153)	(1.000)
Systematic Bias	0.104		0.150	0.200
	(0.006)		(0.063)	(0.134)
Endogenous		0.146	0.201	0.214
		(0.028)	(0.014)	(0.065)
Systematic Bias × Endogenous			-0.037	-0.148
			(0.774)	(0.486)
Observations	930	330	155	55
Dependent variable mean	0.549	0.609	0.232	0.164

Notes: Results from a linear regression with $p-$values in parentheses and standard errors clustered at the officer level. The dependent variable Chose High Rate is a dummy variable that equals 1 if the city with the higher average reporting rate was predicted to have higher crime. The dependent variable Repeat Prediction is a dummy that equals 1 if the city predicted to have higher crime is the same as in the previous prediction. The dependent variable Persistent Prediction is a dummy that equals 1 if an officer predicted the same city to have higher crime across the last six decisions of Part II. Systematic Bias is a dummy variable that equals 1 if the officer was randomized into the *Selection Bias Across Groups* treatment, where areas from one of the two cities have a 30 percentage points higher reporting rate in expectation than the other city. It equals 0 when the officer is in the *Uniform Selection Bias* treatment, where reporting rates for both cities are drawn from the same uniform distribution, and thus are equal in expectation. Endogenous is a dummy variable that equals 1 if an officer is randomized into the *Endogenous Selection* treatment, where predicting one city to have higher crime increases reporting rates in that city by 15 percentage points in the next prediction. It equals 0 when the officer is in the *Exogenous Selection* treatment, where predictions do not affect future reporting rates. Column (1) uses the first six predictions of Part II. Column (2) uses the last six predictions in Part II, only including officers randomized into the Control algorithm.. Column (3) aggregates the last six predictions of Part II at the officer level. Column (4) repeats this analysis but only including officers randomized into the Control algorithm.

Table A.12: Prediction Bias Across Groups, Selection Environments

	Prediction Bias		
	(1)	(2)	(3)
Constant	0.178	0.218	0.208
	(0.000)	(0.000)	(0.000)
Selection Bias	0.093		0.025
	(0.006)		(0.734)
Endogenous		0.104	0.149
		(0.059)	(0.028)
Endogenous × Selection Bias			-0.093
			(0.395)
Observations	155	55	55
Dependent variable mean	0.226	0.273	0.273

Notes: Results from a linear regression with $p-$values in parentheses and standard errors clustered at the officer level. The dependent variable Prediction Bias is the officer-level deviation from the optimal likelihood to predict higher crime in one of the groups. We first calculate the officer-level optimal benchmark: how often their optimal prediction would be city C. Note that, although on expectation both cities have equal crime, because of randomization the benchmark is not 50% for all officers. We then calculate prediction bias as the deviation from this optimal benchmark, in absolute value, for each officer. Selection Bias is a dummy variable that equals 1 if the officer was randomized into the *Selection Bias Across Groups* treatment, where areas from one of the two cities have a 30 percentage points higher reporting rate in expectation than the other city. It equals 0 when the officer is in the *Uniform Selection Bias* treatment, where reporting rates for both cities are drawn from the same uniform distribution, and thus are equal in expectation. Endogenous is a dummy variable that equals 1 if an officer is randomized into the *Endogenous Selection* treatment, where predicting one city to have higher crime increases reporting rates in that city by 15 percentage points in the next prediction. It equals 0 when the officer is in the *Exogenous Selection* treatment, where predictions do not affect future reporting rates. Column (1) uses the first six predictions of Part II. Column (2) and (3) use the last six predictions in Part II, only including officers randomized into the Control algorithm.

Table A.13: Standardized response times by decision features

	Standardized Response Time					
	Part I			Part II		
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	-0.085	0.272	0.192	-0.114	0.298	0.071
	(0.167)	(0.038)	(0.114)	(0.119)	(0.135)	(0.683)
Accurate	0.167	-0.411	-0.481	0.245	-0.468	-0.434
	(0.031)	(0.003)	(0.000)	(0.004)	(0.025)	(0.021)
Selection-Sensitive		-0.400	-0.365		-0.444	-0.317
		(0.003)	(0.003)		(0.032)	(0.080)
Selection-Sensitive × Accurate		0.752	0.713		0.855	0.566
		(0.000)	(0.000)		(0.001)	(0.009)
Use Calculator			1.747			1.323
			(0.000)			(0.000)
Observations	1,240	1,240	1,240	930	930	930
Dependent variable mean	0.000	0.000	0.000	0.000	0.000	0.000

Notes: Results from a linear regression with $p-$values in parentheses and standard errors clustered at the officer level. The dependent variable Standardized Response Time is the z-scored response time for each prediction. Accurate is a dummy that equals 1 if the prediction of which area had more crime was accurate. Selection-Sensitive is a dummy that equals 1 if adjusting for selection is needed to make an accurate prediction. Use Calculator is a dummy that equals 1 if the on-screen calculator was used for that prediction. Columns (1) to (3) use the eight predictions from Part I. Columns (4) to (6) use the first six predictions of Part II.